Which revision fixes "a bright, dark, cheerful, gloomy forest"?

Answer: C. "A bright, cheerful forest in the morning"

Pick one consistent mood.

- ✗ A. "A forest": too vague
- ✗ B. "Any forest you like": too vague
- ✗ D. "A bright, dark, cheerful, gloomy, spooky": more conflict

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-M-Q1 · question page 1514

AI.IMAGE.02 · Inspect a generated image for visual errors and garbled text

Which part of a generated image often has errors?

Answer: D. Hands and fingers

Extra or missing fingers.

- ✗ A. The image size: not a visual flaw
- ✗ B. The overall color of the sky, which image tools always get wrong in every picture: not the common issue
- ✗ C. The file name: not part of the image

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-M-Q2 · question page 1515

AI.IMAGE.02 · Inspect a generated image for visual errors and garbled text

In this prepared example, which word is misspelled?

Prepared example (a written description of a fictional generated image)

Image description: a cheerful flyer with cupcakes. Visible text reads "SPRING BAKE SAEL - Saturday, April 31 - 10 am to 2 pm - Room 12."

Answer: C. SAEL

It should read SALE.

- ✗ A. Room: spelled correctly
- ✗ B. Saturday: spelled correctly
- ✗ D. SPRING: spelled correctly

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-M-Q3 · question page 1516

AI.IMAGE.02 · Inspect a generated image for visual errors and garbled text

In the same example, what other error appears?

Prepared example (a written description of a fictional generated image)

Image description: a cheerful flyer with cupcakes. Visible text reads "SPRING BAKE SAEL - Saturday, April 31 - 10 am to 2 pm - Room 12."

Answer: B. April 31 does not exist

April has 30 days.

- ✗ A. Room 12 is not a real number: it is fine
- ✗ C. Saturday is misspelled: it is correct
- ✗ D. 10 am to 2 pm is too long: not an error

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-M-Q4 · question page 1517

AI.IMAGE.02 · Inspect a generated image for visual errors and garbled text

How should you check text in an image?

Answer: A. Zoom in and read it letter by letter

Careful reading.

- ✗ B. Trust it if the image is pretty: not a check
- ✗ C. Count the colors: not about text
- ✗ D. Glance at it quickly from a distance: misses errors

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-M-Q5 · question page 1518

AI.IMAGE.02 · Inspect a generated image for visual errors and garbled text

What is often easier than fixing text the generator made?

Answer: B. Adding the text yourself in a design app

Add text yourself.

- ✗ A. Leaving the errors in: not good
- ✗ C. Removing all images: unnecessary
- ✗ D. Asking for the same image again and again: unreliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-B1-Q1 · question page 1519

AI.IMAGE.02 · Inspect an image with a checklist

Six fingers on a waving hand fails which point?

Answer: D. Bodies

Count fingers.

- ✗ A. Text: not text
- ✗ B. Patterns: not patterns
- ✗ C. Light: not light

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-B1-Q2 · question page 1520

AI.IMAGE.02 · Inspect an image with a checklist

A cup merging into a table fails which point?

Answer: A. Objects

Blending objects.

- ✗ B. Text: not text
- ✗ C. Light: not light
- ✗ D. Bodies: not stated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-B1-Q3 · question page 1521

AI.IMAGE.02 · Inspect an image with a checklist

A sign reading "Opne" fails which point?

Answer: C. Text

Misspelled word.

- ✗ A. Objects: not objects
- ✗ B. Bodies: not bodies
- ✗ D. Patterns: not patterns

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-B1-Q4 · question page 1522

AI.IMAGE.02 · Inspect an image with a checklist

Shadows falling in two different directions fail which point?

Answer: D. Light

Inconsistent lighting.

✗ A. Text: no

✗ B. Bodies: no

✗ C. Objects: not the category

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-B1-Q5 · question page 1523

AI.IMAGE.02 · Inspect an image with a checklist

A striped shirt whose stripes change direction fails which point?

Answer: C. Patterns

Broken pattern.

✗ A. Bodies: not bodies

✗ B. Light: not light

✗ D. Text: not text

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F1-Q1 · question page 1524

AI.IMAGE.02 · Catch image flaws before using an image

A poster image shows the event phone number with one digit changed. What should you do?

Answer: B. Fix the number before printing

Wrong details cause problems.

✗ A. Remove the event date too: unrelated

✗ C. Make the number smaller: still wrong

✗ D. Print it anyway: wrong contact

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F1-Q2 · question page 1525

AI.IMAGE.02 · Catch image flaws before using an image

A family portrait for a slideshow has a child with three arms. What is the best choice?

Answer: A. Regenerate, edit or choose another image

Do not use flawed images.

- ✗ B. Make it black and white: still wrong
- ✗ C. Keep it, since viewers rarely look closely at people in pictures shown during slideshows: noticeable
- ✗ D. Add a caption explaining it: still odd

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F1-Q3 · question page 1526

AI.IMAGE.02 · Catch image flaws before using an image

A store sign in a generated image reads "Grocey Stroe." What kind of flaw is this?

Answer: B. Garbled text

Misspelled text.

- ✗ A. No flaw: misspelled
- ✗ C. A pattern flaw: not pattern
- ✗ D. A lighting flaw: it is spelling

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F1-Q4 · question page 1527

AI.IMAGE.02 · Catch image flaws before using an image

Before posting a generated image online, what is the best first step?

Answer: C. Zoom in and check details

Inspect first.

✗ A. Post it right away, since generated images are checked for errors before you receive them: not guaranteed

✗ B. Add a filter: not a check

✗ D. Compress it: not a check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F1-Q5 · question page 1528

AI.IMAGE.02 · Catch image flaws before using an image

A flyer has correct text that you added yourself and a generated background. What still needs checking?

Answer: D. The background for visual flaws

Check all generated parts.

✗ A. Only the font: not enough

✗ B. Nothing, because adding text yourself also fixes every flaw in the generated background: not true

✗ C. Only the file size: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F2-Q1 · question page 1529

AI.IMAGE.02 · Prioritize the flaws that matter

Which flaw matters most on an event flyer?

Answer: A. A wrong date

Wrong facts cause real problems.

✗ B. A tree with odd leaves: minor

✗ C. A slightly blurry corner: minor

✗ D. A cloud in the background: harmless

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S73-F2-Q2 · question page 1530

AI.IMAGE.02 · Prioritize the flaws that matter

A certificate image misspells the winner's name. Why is this serious?

Answer: D. It gives wrong information about a real person

Names must be right.

- ✗ A. It changes the paper size: irrelevant
- ✗ B. It is not serious, because names on certificates are decorative and nobody reads them: they matter
- ✗ C. It makes the file larger: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F2-Q3 · question page 1531

AI.IMAGE.02 · Prioritize the flaws that matter

A generated safety sign looks almost like a real one but the symbol is slightly different. What is the risk?

Answer: B. People may misread the warning

Wrong symbols mislead.

- ✗ A. The colors fade: not the risk
- ✗ C. No risk, because people always understand safety signs even when the symbol is changed: not guaranteed
- ✗ D. The sign becomes funny: not the risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F2-Q4 · question page 1532

AI.IMAGE.02 · Prioritize the flaws that matter

An image looks fine on a phone. Why check before printing a large banner?

Answer: A. Flaws can be obvious when enlarged

Size reveals details.

- ✗ B. Because banners never show text: they do
- ✗ C. Because printing always changes every color in a picture into completely different colors: not the point
- ✗ D. Because phones show extra details: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S73-F2-Q5 · question page 1533

AI.IMAGE.02 · Prioritize the flaws that matter

A generated logo looks like another company's logo. What should you do?

Answer: C. Avoid using it

It may belong to someone else.

- ✗ A. Add your name and use it: still similar
- ✗ B. Change one color and use it: still similar
- ✗ D. Use it: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-M-Q1 · question page 1535

AI.IMAGE.03 · Keep a series consistent and request targeted edits

What is drift in an image series?

Answer: C. Details changing from one image to the next

Changing hair, clothes or style.

- ✗ A. Images loading slowly: not the meaning

✗ B. Images getting larger: not the meaning

✗ D. Images slowly moving across the screen: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-M-Q2 · question page 1536

AI.IMAGE.03 · Keep a series consistent and request targeted edits

How can you keep a character consistent?

Answer: A. Reuse the same detailed description each time

Same words, same features.

✗ B. Change the style each page: causes drift

✗ C. Leave out the description: causes drift

✗ D. Describe the character differently each time: causes drift

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-M-Q3 · question page 1537

AI.IMAGE.03 · Keep a series consistent and request targeted edits

Which edit request is targeted?

Answer: D. "Make only the raincoat green; keep everything else."

One clear change.

✗ A. "Make it different.": vague

✗ B. "Change it.": vague

✗ C. "Redraw the whole picture again from the beginning.": not targeted

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-M-Q4 · question page 1538

AI.IMAGE.03 · Keep a series consistent and request targeted edits

After an edit, what should you do?

Answer: B. Compare the result with the original

Compare versions.

- ✗ A. Post it immediately: not checked
- ✗ C. Make five more edits without looking: risky
- ✗ D. Delete the original image right away: loses comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-M-Q5 · question page 1539

AI.IMAGE.03 · Keep a series consistent and request targeted edits

What tool feature can help consistency?

Answer: A. Reference images

Show the tool what to match.

- ✗ B. A faster internet connection: irrelevant
- ✗ C. A larger screen: not a tool feature
- ✗ D. A feature that makes every image: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-B1-Q1 · question page 1540

AI.IMAGE.03 · Compare a series with a school uniform

What does a uniform keep the same?

Answer: D. Colors, clothes and logo

Consistent features.

- ✗ A. The timetable: no

✗ B. The weather: no

✗ C. Each student's height: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-B1-Q2 · question page 1541

AI.IMAGE.03 · Compare a series with a school uniform

Students choosing new outfits daily is like what?

Answer: C. Drift in an image series

Changing details.

✗ A. A targeted edit: no

✗ B. A reference image: no

✗ D. A perfectly consistent series: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-B1-Q3 · question page 1542

AI.IMAGE.03 · Compare a series with a school uniform

A character's "uniform" is what?

Answer: B. The same written description used every time

Reuse the description.

✗ A. A new style each page: drift

✗ C. No description: drift

✗ D. A different description for every page: causes drift

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-B1-Q4 · question page 1543

AI.IMAGE.03 · Compare a series with a school uniform

Why do mismatched photos look odd?

Answer: A. Things that should match do not

Inconsistency is noticeable.

- ✗ B. Because uniforms are blue: irrelevant
- ✗ C. Because photos taken on different days always come out blurry and unusable: not the point
- ✗ D. Because schools are large: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-B1-Q5 · question page 1544

AI.IMAGE.03 · Compare a series with a school uniform

What is the lesson for an image series?

Answer: D. Keep the same details each time

Consistency.

- ✗ A. Use only one image: not the lesson
- ✗ B. Avoid characters: not the lesson
- ✗ C. Change everything each time: confusing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F1-Q1 · question page 1545

AI.IMAGE.03 · Keep series consistent and edit precisely

A children's story has a dog that is brown on page 1 and gray on page 3. What should the author do?

Answer: C. Reuse one exact description of the dog for every page

A fixed description.

✗ A. Use gray on every page without saying so: still drift

✗ B. Remove the dog: unnecessary

✗ D. Change the story so that the dog changes color by magic: unnecessary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F1-Q2 · question page 1546

AI.IMAGE.03 · Keep series consistent and edit precisely

Nia wants to change only the background of an image from a park to a beach. Which request fits?

Answer: B. "Change only the background to a beach; keep the people the same."

Targeted edit.

✗ A. "Beach.": vague

✗ C. "Change everything to look like summer.": too broad

✗ D. "Make a new beach picture with some people in it who look somewhat like the old ones.": not targeted

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F1-Q3 · question page 1547

AI.IMAGE.03 · Keep series consistent and edit precisely

A slide series uses watercolor on some slides and photos on others. What should change?

Answer: A. Use the same style words on every slide

Consistent style words.

✗ B. Use random styles: less consistent

✗ C. Add more styles, so the slides look varied and keep the audience's attention throughout: less consistent

✗ D. Remove all images: unnecessary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F1-Q4 · question page 1548

AI.IMAGE.03 · Keep series consistent and edit precisely

After asking to change a shirt to blue, what should you check?

Answer: C. That only the shirt changed

Compare with the original.

- ✗ A. Only the file name: not enough
- ✗ B. Nothing, because a targeted edit request always changes exactly one thing in any picture: not guaranteed
- ✗ D. Only the size: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F1-Q5 · question page 1549

AI.IMAGE.03 · Keep series consistent and edit precisely

Which description helps a character stay consistent?

Answer: B. "Leo: tall, red cap, green scarf, freckles."

Specific, reusable features.

- ✗ A. "A boy.": too vague
- ✗ C. "Leo.": too vague
- ✗ D. "Leo: a boy, and he looks however you think is best for the story on each page.": invites drift

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F2-Q1 · question page 1550

AI.IMAGE.03 · Catch edits that change too much

You asked to change the sky, and the person's face also changed. What does this show?

Answer: D. Edits can change more than asked

Check the whole image.

- ✗ A. That the sky and face are the same: no
- ✗ B. That edits never work: too strong
- ✗ C. That the tool always changes faces: not on purpose

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F2-Q2 · question page 1551

AI.IMAGE.03 · Catch edits that change too much

Why keep a copy of each version during edits?

Answer: A. So you can go back if an edit goes wrong

Undo.

- ✗ B. So images look brighter: not the reason
- ✗ C. To use more storage: not the reason
- ✗ D. Because tools refuse to edit any image: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F2-Q3 · question page 1552

AI.IMAGE.03 · Catch edits that change too much

What is a style sheet for an image series?

Answer: B. A list of each character's features, colors and style

A reference for consistency.

- ✗ A. A list of fonts only: too narrow
- ✗ C. A spreadsheet of prices: no

✗ D. A document that lists the printer settings for every page: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F2-Q4 · question page 1553

AI.IMAGE.03 · Catch edits that change too much

A new image almost matches, but the character's glasses are missing. What should you do?

Answer: C. Reject or fix it using the style sheet

Check against the style sheet.

✗ A. Remove glasses from all other images: more work
✗ B. Change the character's name: irrelevant
✗ D. Accept it: readers notice

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S74-F2-Q5 · question page 1554

AI.IMAGE.03 · Catch edits that change too much

When should you check a new image against the style sheet?

Answer: D. Before accepting it

Check each image.

✗ A. After the whole series is printed and handed out, so you can fix it next time: too late
✗ B. Only for the first image: not enough
✗ C. Never: loses consistency

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-M-Q1 · question page 1556

AI.IMAGE.04 · Check generated diagrams, maps and charts against real data

Do image generators check facts when they draw a map?

Answer: A. No; they make pictures that look like maps

Look, not accuracy.

- ✗ B. Yes; every generated map is checked against official satellite data before it is shown: not true
- ✗ C. Only for small countries: no
- ✗ D. Only in color: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-M-Q2 · question page 1557

AI.IMAGE.04 · Check generated diagrams, maps and charts against real data

In this prepared example, which bar is drawn wrongly?

Prepared example (a written description of a fictional generated bar chart)

Bar label	Value printed on bar	Bar height drawn (grid squares)
Monday	10	1
Tuesday	40	4
Wednesday	20	5

Answer: B. Wednesday

At one square per 10, Wednesday (20) should be 2 squares, not 5.

- ✗ A. Tuesday: 40 drawn as 4 squares matches
- ✗ C. None: heights do not match
- ✗ D. Monday: 10 as 1 square matches

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-M-Q3 · question page 1558

AI.IMAGE.04 · Check generated diagrams, maps and charts against real data

How should you make a chart meant to inform?

Answer: C. Build it from real data, such as in a spreadsheet

Use real data.

- ✗ A. Copy a chart from memory: unreliable
- ✗ B. Guess the numbers: unreliable
- ✗ D. Ask an image generator to draw a chart that looks professional and has lots of colors: no real data

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-M-Q4 · question page 1559

AI.IMAGE.04 · Check generated diagrams, maps and charts against real data

What should generated images be used for?

Answer: D. Decoration, not evidence

Illustration only.

- ✗ A. Medical diagrams for patients: not reliable
- ✗ B. Official maps: not reliable
- ✗ C. Evidence in reports: not accurate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-M-Q5 · question page 1560

AI.IMAGE.04 · Check generated diagrams, maps and charts against real data

A generated science diagram labels the heart as the lungs. What does this show?

Answer: B. Labels can be wrong

Check every label.

- ✗ A. That the tool is always right: not true
- ✗ C. That diagrams cannot have labels: they can
- ✗ D. That the heart and lungs are the same organ in some science textbooks used in schools: false

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-B1-Q1 · question page 1561

AI.IMAGE.04 · Check a chart against its numbers

A = 5, B = 10, C = 15. Which bar should be tallest?

Answer: C. C

Biggest number, tallest bar.

- ✗ A. B: middle value
- ✗ B. A, because the first bar in a chart is always drawn as the tallest one: no
- ✗ D. All equal: values differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-B1-Q2 · question page 1562

AI.IMAGE.04 · Check a chart against its numbers

Which should be shortest?

Answer: A. A

Smallest number.

- ✗ B. B: middle
- ✗ C. C, since the last bar in any chart is usually drawn the shortest of all: no
- ✗ D. None: one is smallest

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S75-B1-Q3 · question page 1563

AI.IMAGE.04 · Check a chart against its numbers

B = 10 and A = 5. How should B compare with A?

Answer: D. About twice as tall

10 is twice 5.

- ✗ A. Half as tall: reversed
- ✗ B. Ten times taller: too much
- ✗ C. About the same height: not proportional

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S75-B1-Q4 · question page 1564

AI.IMAGE.04 · Check a chart against its numbers

B is drawn shorter than A. What does that mean?

Answer: A. The chart is wrong

Heights must match numbers.

- ✗ B. A must be 20: label says 5
- ✗ C. B must be 3: label says 10
- ✗ D. The chart is correct: misleading

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-B1-Q5 · question page 1565

AI.IMAGE.04 · Check a chart against its numbers

What else should you read?

Answer: C. Whether the axis starts at 0

Axes can mislead.

✗ A. The chart's font: not the check

✗ B. The file name: not the check

✗ D. The color of each bar, because colors show the real values more clearly than heights: not the check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F1-Q1 · question page 1566

AI.IMAGE.04 · Use diagrams and charts responsibly in practice

Ana needs a map of bus routes for new students. What is the best source?

Answer: B. The transit agency's official map

Official, accurate source.

✗ A. An image generator asked to draw a detailed and accurate map of the city's bus routes: may invent routes

✗ C. A map of another city: wrong

✗ D. A sketch from memory: unreliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F1-Q2 · question page 1567

AI.IMAGE.04 · Use diagrams and charts responsibly in practice

A generated chart for a club report shows membership numbers. What should be checked?

Answer: D. Every number against club records

Compare with real data.

✗ A. Nothing: must check

✗ B. Only that the colors match the club logo, since the numbers come from the tool and are fine: numbers may be wrong

✗ C. Only the title: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F1-Q3 · question page 1568

AI.IMAGE.04 · Use diagrams and charts responsibly in practice

A teacher wants a pretty background picture of planets for a slide title. Is a generated image reasonable?

Answer: B. Yes, as decoration labeled as an illustration

Decoration is fine.

✗ A. No, because generated images can never be used for anything at all in any classroom setting: too strong
✗ C. Only if it shows text: irrelevant
✗ D. Yes, and it can be used as a scientific diagram too: not accurate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F1-Q4 · question page 1569

AI.IMAGE.04 · Use diagrams and charts responsibly in practice

Which is the most accurate way to make a pie chart of a survey?

Answer: C. Enter the survey results into a spreadsheet chart

Real data in.

✗ A. Describe the survey to an image tool and ask for a pie chart that looks about right: not accurate
✗ B. Use a chart from another survey: wrong data
✗ D. Draw slices by guessing: not accurate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F1-Q5 · question page 1570

AI.IMAGE.04 · Use diagrams and charts responsibly in practice

A generated diagram of a plant has a part labeled "photosynthesis root." What should you do?

Answer: A. Check labels against a trusted source

Verify labels.

- ✗ B. Add more labels: more errors
- ✗ C. Make it bigger: not a fix
- ✗ D. Keep it, since diagrams always use correct scientific names even when they look unusual: not guaranteed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F2-Q1 · question page 1571

AI.IMAGE.04 · Spot subtle chart and diagram errors

A chart's axis starts at 50, so 52 looks twice as tall as 51. What is the problem?

Answer: D. Small differences look huge

Truncated axes exaggerate.

- ✗ A. The numbers are wrong: the scale misleads
- ✗ B. The chart is too colorful: not the problem
- ✗ C. No problem: misleading

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F2-Q2 · question page 1572

AI.IMAGE.04 · Spot subtle chart and diagram errors

A diagram is right except two labels are swapped. Why is it still a problem?

Answer: C. Readers learn the wrong names

Partly wrong is still wrong.

- ✗ A. Because diagrams need three labels: no
- ✗ B. Because swapped labels change the colors: no
- ✗ D. It is not a problem: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F2-Q3 · question page 1573

AI.IMAGE.04 · Spot subtle chart and diagram errors

A map has the right shape but the wrong scale. What could go wrong?

Answer: A. Distances would be misjudged

Scale affects distance.

- ✗ B. Street names would change: not the issue
- ✗ C. The colors would fade: irrelevant
- ✗ D. Nothing: scale matters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F2-Q4 · question page 1574

AI.IMAGE.04 · Spot subtle chart and diagram errors

Why label a generated picture as "illustration" next to real data?

Answer: B. So readers do not treat it as accurate

Clear labeling.

- ✗ A. Because labels are required by every printer: no
- ✗ C. To make the file smaller: irrelevant
- ✗ D. So readers know the illustration is more accurate: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S75-F2-Q5 · question page 1575

AI.IMAGE.04 · Spot subtle chart and diagram errors

Which habit catches most chart errors?

Answer: D. Reading axes and legends and comparing with the source

Careful reading and comparison.

- ✗ A. Checking the font: not enough
- ✗ B. Counting the colors: not enough
- ✗ C. Looking at the overall shape of the chart for a second: misses errors

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-M-Q1 · question page 1577

AI.IMAGE.05 · Plan a presentation and tell a text draft from creating a file

What does a clear presentation structure include?

Answer: A. A purpose, main points with evidence, and a close

Purpose, points, evidence, close.

- ✗ B. Only pictures: missing structure
- ✗ C. Only a title slide: missing structure
- ✗ D. As many slides as possible, each packed with full paragraphs of text so nothing is left out: too much

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-M-Q2 · question page 1578

AI.IMAGE.05 · Plan a presentation and tell a text draft from creating a file

In this fictional example, what did the tool produce?

Fictional example (prepared in advance; not a real AI output)

Request:

Make me a 5-slide presentation on recycling.

Response:

Slide 1: Why Recycling Matters
Slide 2: What Can Be Recycled
Slide 3: How Our School Recycles
Slide 4: Common Mistakes
Slide 5: What You Can Do
(Each slide has 3 bullet points and speaker notes below.)

Answer: C. A text outline of slides

Text describing slides.

✗ A. A finished slide file saved on the computer that is ready to open in any presentation app: only text

✗ B. A video: no

✗ D. A printed handout: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-M-Q3 · question page 1579

AI.IMAGE.05 · Plan a presentation and tell a text draft from creating a file

How is an actual slide file usually made?

Answer: D. In a presentation app or with a real file-making feature

File creation is a separate step.

✗ A. By typing the word "slides" into a chat, which always creates a real file automatically: not always

✗ B. By printing the chat: not a slide file

✗ C. By renaming a text file: not a slide file

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-M-Q4 · question page 1580

AI.IMAGE.05 · Plan a presentation and tell a text draft from creating a file

What is a common rule for slides?

Answer: B. One idea per slide, few words

Simple, focused slides.

- ✗ A. Use full paragraphs: too wordy
- ✗ C. No titles: unclear
- ✗ D. Put every idea from the whole talk: crowded

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-M-Q5 · question page 1581

AI.IMAGE.05 · Plan a presentation and tell a text draft from creating a file

After a slide file is created, what should you do?

Answer: D. Open it and check every slide

Check the real file.

- ✗ A. Delete the outline: not needed
- ✗ B. Email it unopened: risky
- ✗ C. Present it straight away: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-B1-Q1 · question page 1582

AI.IMAGE.05 · Compare a text draft with a recipe versus a cooked meal

A text draft of slides is like what?

Answer: B. A recipe

Instructions, not the result.

- ✗ A. A grocery store: no
- ✗ C. A kitchen: no
- ✗ D. A cooked meal: not yet made

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-B1-Q2 · question page 1583

AI.IMAGE.05 · Compare a text draft with a recipe versus a cooked meal

The slide file is like what?

Answer: A. The cooked meal

The finished product.

- ✗ B. The oven: no
- ✗ C. The shopping list: no
- ✗ D. The recipe card: that is the draft

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-B1-Q3 · question page 1584

AI.IMAGE.05 · Compare a text draft with a recipe versus a cooked meal

Where is the meal made?

Answer: C. In a kitchen

Like a presentation app.

- ✗ A. At the table: no
- ✗ B. Inside the recipe book itself, as soon as someone reads the recipe out loud: no
- ✗ D. In the store: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-B1-Q4 · question page 1585

AI.IMAGE.05 · Compare a text draft with a recipe versus a cooked meal

Tasting the meal before serving is like what?

Answer: B. Opening and checking the file

Check the real result.

- ✗ A. Washing dishes: no
- ✗ C. Buying more food: no
- ✗ D. Reading the recipe one more time: not checking the result

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-B1-Q5 · question page 1586

AI.IMAGE.05 · Compare a text draft with a recipe versus a cooked meal

Can you present a recipe as if it were a meal?

Answer: C. No; it must be made first

Draft is not file.

- ✗ A. Only on holidays: no
- ✗ B. Yes, if it is long: no
- ✗ D. Yes, because a recipe and a meal are exactly the same thing when guests arrive: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F1-Q1 · question page 1587

AI.IMAGE.05 · Plan and produce presentations in practice

Sam asks a chat for "slides" and gets bullet points in the chat window. What should he do next?

Answer: D. Build the slides in a presentation app from the text

Turn the draft into a file.

- ✗ A. Present the chat window: not slides
- ✗ B. Give up: unnecessary
- ✗ C. Look for the slide file in his downloads folder: may not exist

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F1-Q2 · question page 1588

AI.IMAGE.05 · Plan and produce presentations in practice

A 10-minute talk has a 30-slide draft. What should change?

Answer: A. Cut it to about 5 to 10 slides

About one slide per minute or two.

- ✗ B. Remove the closing slide only: not enough
- ✗ C. Add 10 more slides: worse
- ✗ D. Keep all 30 slides: too many

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F1-Q3 · question page 1589

AI.IMAGE.05 · Plan and produce presentations in practice

Which slide follows the "few words" rule?

Answer: D. Title "Recycling Saves Energy" with one chart

One idea, clear visual.

- ✗ A. A slide with 12 bullet points: too many
- ✗ B. A slide with only tiny text: hard to read
- ✗ C. A slide with four full paragraphs explaining every detail of recycling from start to finish: too wordy

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F1-Q4 · question page 1590

AI.IMAGE.05 · Plan and produce presentations in practice

A tool with a real file feature makes a slide file. What should you check?

Answer: C. Every slide in the opened file

Check all slides.

- ✗ A. Only the first slide, because if the first one is fine then every other slide is fine too: not enough
- ✗ B. Nothing: must check
- ✗ D. Only the file name: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F1-Q5 · question page 1591

AI.IMAGE.05 · Plan and produce presentations in practice

Which use of AI fits presentation planning best?

Answer: A. Drafting an outline you then revise

A starting point.

- ✗ B. Skipping preparation: not good
- ✗ C. Reading the AI text word for word: not yours
- ✗ D. Letting the tool decide the purpose: you decide

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F2-Q1 · question page 1592

AI.IMAGE.05 · Check exported slides and make the talk your own

Slides look fine on a laptop but text overflows on the classroom screen. What should have

happened?

Answer: B. Checking on the device that will show them

Test on the target device.

- ✗ A. Nothing, since every screen shows slides in exactly the same way regardless of the computer: they differ
- ✗ C. Using smaller images: not the issue
- ✗ D. Using more text: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F2-Q2 · question page 1593

AI.IMAGE.05 · Check exported slides and make the talk your own

Speaker notes say "studies show 90% of waste is recyclable." What should you do?

Answer: C. Check the claim before saying it

Notes are unverified answers.

- ✗ A. Delete all notes: unnecessary
- ✗ B. Make the number bigger: no
- ✗ D. Say it confidently, since speaker notes are written from verified sources by the tool: not guaranteed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F2-Q3 · question page 1594

AI.IMAGE.05 · Check exported slides and make the talk your own

An exported file replaced your font and lost an image. What kind of problem is this?

Answer: D. An export problem

Exports can change files.

✗ A. A spelling problem: no

✗ B. A grounding problem: no

✗ C. A problem with the topic: not the topic

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F2-Q4 · question page 1595

AI.IMAGE.05 · Check exported slides and make the talk your own

Why reorder and rewrite a generated outline?

Answer: A. So it fits your audience and time

Make it yours.

✗ B. So it becomes longer: not the goal

✗ C. Because teachers require rewriting: not the main reason

✗ D. Because generated outlines always contain nothing useful and must be thrown away: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S76-F2-Q5 · question page 1596

AI.IMAGE.05 · Check exported slides and make the talk your own

About how many slides suit a 20-minute talk?

Answer: B. About 10 to 20

One slide per minute or two.

✗ A. Exactly 1: too few

✗ C. About 200: far too many

✗ D. About 80: too many

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S77-M-Q1 · question page 1598

AI.IMAGE.06 · Make visuals accessible with alt text, contrast and labels

What is alt text?

Answer: A. A short description so screen readers can read an image aloud

A description for accessibility.

- ✗ B. A caption in bold: not the same
- ✗ C. A hidden password: no
- ✗ D. The file name of the image, such as "photo123.jpg,": not useful

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-M-Q2 · question page 1599

AI.IMAGE.06 · Make visuals accessible with alt text, contrast and labels

Which alt text is best for a sales chart?

Answer: B. "Bar chart: sales doubled from March to June."

States what matters.

- ✗ A. "chart.png": useless
- ✗ C. "Image": useless
- ✗ D. "A chart image that someone made for this report.": misses the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-M-Q3 · question page 1600

AI.IMAGE.06 · Make visuals accessible with alt text, contrast and labels

What does good contrast mean?

Answer: D. Text stands out clearly from its background

Easy-to-see text.

✗ A. Text is very small: not contrast

✗ B. Text is in all capitals: not contrast

✗ C. Text uses many different bright colors: poor contrast

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-M-Q4 · question page 1601

AI.IMAGE.06 · Make visuals accessible with alt text, contrast and labels

Why put labels on chart lines instead of relying on color?

Answer: C. Some viewers cannot tell the colors apart

Color alone fails some people.

✗ A. To hide the data: no

✗ B. To make the chart smaller: not the reason

✗ D. Because colors are not allowed in any chart: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-M-Q5 · question page 1602

AI.IMAGE.06 · Make visuals accessible with alt text, contrast and labels

Can AI draft alt text?

Answer: A. Yes, but you check it

Draft, then verify.

✗ B. No, it cannot describe images: it can

✗ C. Yes, and its drafts never need checking, since tools describe images perfectly every time: they can be wrong

✗ D. Only for photos of food: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-B1-Q1 · question page 1603

AI.IMAGE.06 · Compare alt text with describing a picture over the phone

Saying "It's a file called IMG_204" is like what?

Answer: B. Useless alt text

Not a description.

- ✗ A. A chart label: no
- ✗ C. Good contrast: no
- ✗ D. Excellent alt text: not useful

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-B1-Q2 · question page 1604

AI.IMAGE.06 · Compare alt text with describing a picture over the phone

"Our team holding the trophy after the final" is like what?

Answer: C. Good alt text

What matters, briefly.

- ✗ A. Poor contrast: no
- ✗ B. A caption in bold: not the point
- ✗ D. A file name: it is a description

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-B1-Q3 · question page 1605

AI.IMAGE.06 · Compare alt text with describing a picture over the phone

Why skip the color of the wall?

Answer: D. It is not what matters

Focus on the purpose.

✗ A. Because walls are invisible: no

✗ B. To make it longer: no

✗ C. Because colors must never be mentioned in any description of any image at all: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-B1-Q4 · question page 1606

AI.IMAGE.06 · Compare alt text with describing a picture over the phone

Who is alt text for?

Answer: B. People who cannot see the image

Screen reader users and others.

✗ A. Search engines only: not the main purpose

✗ C. Printers: no

✗ D. Only the person who created the image: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-B1-Q5 · question page 1607

AI.IMAGE.06 · Compare alt text with describing a picture over the phone

A good phone description is...

Answer: D. Short and focused on what matters

Brief and meaningful.

✗ A. A file name: useless

✗ B. Only one word: too little

✗ C. Very long, listing every single thing that appears anywhere in the picture: too much

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F1-Q1 · question page 1608

AI.IMAGE.06 · Make visuals accessible in practice

Kim's slide uses yellow text on a white background. What should she change?

Answer: A. Use dark text on the light background

Improve contrast.

- ✗ B. Remove the text: loses content
- ✗ C. Add more yellow: worse
- ✗ D. Make the yellow text slightly larger: contrast still poor

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F1-Q2 · question page 1609

AI.IMAGE.06 · Make visuals accessible in practice

A line chart uses red and green lines with no labels. What is the best fix?

Answer: C. Label each line with words

Labels beyond color.

- ✗ A. Use two slightly different shades of green for the lines so the chart looks more modern: worse
- ✗ B. Remove the legend: worse
- ✗ D. Make the chart smaller: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F1-Q3 · question page 1610

AI.IMAGE.06 · Make visuals accessible in practice

A tool drafts alt text "A group of people" for a photo of the science fair winners. What should you do?

Answer: A. Revise it to say who and why it matters

Check and improve drafts.

- ✗ B. Add the file name: not useful
- ✗ C. Keep it, since every description written by a tool is already as detailed as needed: misses the point
- ✗ D. Delete the alt text: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F1-Q4 · question page 1611

AI.IMAGE.06 · Make visuals accessible in practice

A newsletter image announces the event date only inside the picture. What helps screen reader users?

Answer: C. Also give the date as real text

Real text is readable.

- ✗ A. Move the picture lower: not a fix
- ✗ B. Add more pictures: not a fix
- ✗ D. Make the date in the picture bigger: they cannot read it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F1-Q5 · question page 1612

AI.IMAGE.06 · Make visuals accessible in practice

Which alt text suits a decorative divider line?

Answer: B. Little or none, since it carries no information

Match alt text to purpose.

- ✗ A. "Important image": misleading
- ✗ C. The file name: not useful

✗ D. A detailed paragraph describing the line's exact color: unneeded

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F2-Q1 · question page 1613

AI.IMAGE.06 · Fix weak alt text and color-only signals

Generated alt text for a chart says "colorful shapes." What is missing?

Answer: D. The data or key finding

Charts need their message.

✗ A. The file size: irrelevant

✗ B. The font name: irrelevant

✗ C. More colors, since describing the colors fully is the most important part of chart alt text: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F2-Q2 · question page 1614

AI.IMAGE.06 · Fix weak alt text and color-only signals

A schedule marks late buses in red only. What improves it?

Answer: A. Add the word "late" or a symbol

Words or symbols, not only color.

✗ B. Remove the late buses: loses information

✗ C. Use pink instead: still color-only

✗ D. Make the red brighter, which ensures every reader can see that a bus is running late: still color-only

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F2-Q3 · question page 1615

AI.IMAGE.06 · Fix weak alt text and color-only signals

Generated alt text mentions "a dog in the corner," but there is no dog. What happened?

Answer: C. The tool guessed something not there

Check drafts against the image.

- ✗ A. Alt text always includes animals: no
- ✗ B. The image changed: unlikely
- ✗ D. The dog must be hidden somewhere: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F2-Q4 · question page 1616

AI.IMAGE.06 · Fix weak alt text and color-only signals

Can screen readers read words that are only part of an image?

Answer: D. No, unless also given as real text

Provide real text.

- ✗ A. Only if the words are in color: no
- ✗ B. Only on weekends: no
- ✗ C. Yes: not reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S77-F2-Q5 · question page 1617

AI.IMAGE.06 · Fix weak alt text and color-only signals

What should alt text match?

Answer: B. The image's purpose

Purpose decides what to say.

- ✗ A. The image size: no

✗ C. The file name: no

✗ D. The length of the page: not the rule

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-M-Q1 · question page 1619

AI.MEDIA.01 · Tell transcription from speech generation and check transcripts

What does transcription do?

Answer: B. Turns spoken audio into written words

Speech to text.

✗ A. Deletes background noise only: not the definition

✗ C. Translates text into music: no

✗ D. Turns written words into a spoken voice: that is speech generation

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-M-Q2 · question page 1620

AI.MEDIA.01 · Tell transcription from speech generation and check transcripts

What does text-to-speech do?

Answer: D. Turns written words into spoken audio

Text to speech.

✗ A. Makes video from text: not this task

✗ B. Checks spelling: no

✗ C. Writes down what people say in a meeting: that is transcription

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-M-Q3 · question page 1621

AI.MEDIA.01 · Tell transcription from speech generation and check transcripts

What is a speaker label?

Answer: A. A note saying who said each part

Who said what.

- ✗ B. The file name: no
- ✗ C. The volume level: no
- ✗ D. A sticker on a loudspeaker: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-M-Q4 · question page 1622

AI.MEDIA.01 · Tell transcription from speech generation and check transcripts

In this prepared example, which detail is most important to check against the recording?

Prepared example (a fictional meeting transcript)

Speaker 1: Our budget for the trip is fifteen hundred dollars.
Speaker 2: We can't book the bus until Friday.
Speaker 1: Then let's ask Mr. Patel to call the company.

Answer: C. "Fifteen hundred dollars" and "can't"

Numbers and negatives change meaning.

- ✗ A. The punctuation only: low stakes
- ✗ B. The word "Then," because short words at the start of sentences are the most often wrong: low stakes
- ✗ D. The word "the": low stakes

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-M-Q5 · question page 1623

AI.MEDIA.01 · Tell transcription from speech generation and check transcripts

"Wreck a nice beach" for "recognize speech" is what kind of error?

Answer: A. A misheard phrase

Similar sounds, wrong words.

- ✗ B. A missing word: not missing
- ✗ C. A text-to-speech error: it is transcription
- ✗ D. A speaker label error: not labeling

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-B1-Q1 · question page 1624

AI.MEDIA.01 · Compare transcription and speech generation with a court reporter and a narrator

A court reporter is like which task?

Answer: C. Transcription

Sound in, text out.

- ✗ A. Translation: no
- ✗ B. Image generation: no
- ✗ D. Speech generation: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-B1-Q2 · question page 1625

AI.MEDIA.01 · Compare transcription and speech generation with a court reporter and a narrator

An audiobook narrator is like which task?

Answer: D. Speech generation

Text in, sound out.

- ✗ A. Editing: no

✗ B. Summarizing: no

✗ C. Transcription: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-B1-Q3 · question page 1626

AI.MEDIA.01 · Compare transcription and speech generation with a court reporter and a narrator

What can a court reporter get wrong?

Answer: B. A name or who spoke

Like transcript errors.

✗ A. Nothing: they can err

✗ C. The judge's salary: irrelevant

✗ D. The color of the courtroom walls, which must be included in every court record: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-B1-Q4 · question page 1627

AI.MEDIA.01 · Compare transcription and speech generation with a court reporter and a narrator

For transcription, what goes in?

Answer: C. Sound

Audio input.

✗ A. Numbers only: no

✗ B. Pictures: no

✗ D. Written text: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-B1-Q5 · question page 1628

AI.MEDIA.01 · Compare transcription and speech generation with a court reporter and a narrator

For speech generation, what comes out?

Answer: D. Sound

Audio output.

- ✗ A. Written text that matches the words someone said in a recorded meeting earlier: reversed
- ✗ B. A spreadsheet: no
- ✗ C. A picture: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F1-Q1 · question page 1629

AI.MEDIA.01 · Use and check transcripts in practice

Jo wants captions for a class video. Which task does she need?

Answer: B. Transcription

Speech becomes text.

- ✗ A. Image generation: no
- ✗ C. Music generation: no
- ✗ D. Text-to-speech: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F1-Q2 · question page 1630

AI.MEDIA.01 · Use and check transcripts in practice

A blind student wants a news article read aloud. Which task fits?

Answer: A. Text-to-speech

Text becomes speech.

✗ B. Image tagging: no

✗ C. Video editing: no

✗ D. Transcription: it is already text

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F1-Q3 · question page 1631

AI.MEDIA.01 · Use and check transcripts in practice

A transcript says the meeting agreed to "cancel" the event, but you recall "continue." What should you do?

Answer: C. Listen to that part of the recording

Check key statements.

✗ A. Ask attendees to vote again: check first

✗ B. Trust the transcript, since automatic transcripts never confuse words that sound alike: they can

✗ D. Delete the recording: loses evidence

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F1-Q4 · question page 1632

AI.MEDIA.01 · Use and check transcripts in practice

A transcript credits the coach with a parent's question. What kind of error is this?

Answer: A. A wrong speaker label

Who said what.

✗ B. A missing word: no

✗ C. A speech generation error: no

✗ D. A misheard word: words fine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F1-Q5 · question page 1633

AI.MEDIA.01 · Use and check transcripts in practice

Which transcript items most need checking?

Answer: B. Names, numbers and decisions

High-stakes details.

- ✗ A. Greetings: low stakes
- ✗ C. Background music: not text
- ✗ D. Filler words like "um" and "uh,": low stakes

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F2-Q1 · question page 1634

AI.MEDIA.01 · Anticipate when transcripts fail

Which recording is most likely to produce transcript errors?

Answer: D. A noisy room with people talking over each other

Noise and crosstalk.

- ✗ A. A short clear voicemail: easier
- ✗ B. A slow, scripted reading: easier
- ✗ C. A quiet room with one person speaking slowly: easiest case

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F2-Q2 · question page 1635

AI.MEDIA.01 · Anticipate when transcripts fail

Why can "can" and "can't" be risky in transcripts?

Answer: B. They sound alike but mean the opposite

Small sound, big meaning.

✗ A. Because transcripts are not allowed to contain any short words with apostrophes: not the reason

✗ C. Because they are rare words: common

✗ D. Because they are spelled the same: they differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F2-Q3 · question page 1636

AI.MEDIA.01 · Anticipate when transcripts fail

A transcript of a medical appointment lists a medicine dose. What should you do?

Answer: D. Confirm it with the provider or the recording

High stakes need confirmation.

✗ A. Double the dose to be safe: dangerous

✗ B. Follow the dose as written, since transcripts of appointments are checked by doctors: not guaranteed

✗ C. Ignore the dose: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F2-Q4 · question page 1637

AI.MEDIA.01 · Anticipate when transcripts fail

Before recording a group conversation, what should you do?

Answer: A. Ask the people for permission

Consent rules vary by place.

✗ B. Record only the loud people: no

✗ C. Use a hidden phone: no

✗ D. Record secretly: rules differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S78-F2-Q5 · question page 1638

AI.MEDIA.01 · Anticipate when transcripts fail

A transcript misspells technical terms in a science lecture. Why?

Answer: C. Technical words are harder to recognize

Unfamiliar vocabulary.

- ✗ A. Because the file was too small: not the reason
- ✗ B. Because the lecturer must have spelled the words incorrectly while speaking them aloud: speech has no spelling
- ✗ D. Because science is not allowed: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-M-Q1 · question page 1640

AI.MEDIA.02 · Check subtitles and translated audio for timing, meaning and names

What are subtitles?

Answer: A. Spoken words shown as text on screen

On-screen text.

- ✗ B. Music under a video: no
- ✗ C. A voice in a second language that replaces the speaker's voice in the original video: that is dubbing
- ✗ D. The video's title only: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-M-Q2 · question page 1641

AI.MEDIA.02 · Check subtitles and translated audio for timing, meaning and names

Which is a timing problem?

Answer: C. A line appears after the person has finished speaking

Out of sync.

- ✗ A. A wrong place name: names
- ✗ B. A misspelled name: names
- ✗ D. A line that uses a word in the wrong language: meaning problem

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-M-Q3 · question page 1642

AI.MEDIA.02 · Check subtitles and translated audio for timing, meaning and names

In this prepared example, which subtitle has a meaning problem?

Prepared example (fictional generated subtitles)

Time	Original (English)	Generated subtitle (Spanish)
0:05	Break a leg tonight!	¡Rómpete una pierna esta noche!
0:09	The show starts at 7.	El espectáculo empieza a las 7.

Answer: B. The 0:05 line

The idiom was translated word for word.

- ✗ A. Neither line: one is literal
- ✗ C. Both lines: only one
- ✗ D. The 0:09 line: it is fine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-M-Q4 · question page 1643

AI.MEDIA.02 · Check subtitles and translated audio for timing, meaning and names

Why can a fluent-sounding dub still be a problem?

Answer: D. It can change the meaning

Fluent is not faithful.

- ✗ A. Because dubs remove the video: no
- ✗ B. Because dubs are always too loud: not the issue
- ✗ C. Because fluent speech is always a sign: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-M-Q5 · question page 1644

AI.MEDIA.02 · Check subtitles and translated audio for timing, meaning and names

Who should review important translated media?

Answer: C. A speaker of both languages

Bilingual review.

- ✗ A. Nobody, since translation tools are now as accurate as professional human translators in every case: not guaranteed
- ✗ B. The video's editor only: may not know both
- ✗ D. Someone who speaks only one language: cannot compare

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-B1-Q1 · question page 1645

AI.MEDIA.02 · Check a subtitle line with three questions

The subtitle appears two seconds late. Which check fails?

Answer: A. Timing

Out of sync.

- ✗ B. None: timing fails
- ✗ C. Names: not names
- ✗ D. Meaning: separate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-B1-Q2 · question page 1646

AI.MEDIA.02 · Check a subtitle line with three questions

"Many animals are falling" fails which check?

Answer: D. Meaning

Idiom taken literally.

- ✗ A. None: meaning fails
- ✗ B. Names: not names
- ✗ C. Timing: not timing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-B1-Q3 · question page 1647

AI.MEDIA.02 · Check a subtitle line with three questions

"Dr. Wen" for "Dr. Nguyen" fails which check?

Answer: B. Names

Wrong name.

- ✗ A. None: names fail
- ✗ C. Timing: not timing
- ✗ D. Meaning: not meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-B1-Q4 · question page 1648

AI.MEDIA.02 · Check a subtitle line with three questions

How many checks fail in the example?

Answer: D. Three

Timing, meaning and names.

- ✗ A. Two: all three
- ✗ B. None: all three
- ✗ C. One: all fail

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-B1-Q5 · question page 1649

AI.MEDIA.02 · Check a subtitle line with three questions

What should "raining cats and dogs" become?

Answer: A. A phrase meaning "raining very hard"

Translate the meaning.

- ✗ B. Nothing at all: loses meaning
- ✗ C. "It is sunny": wrong meaning
- ✗ D. A word-for-word version that keeps the cats and the dogs, so the subtitle is exact: loses meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F1-Q1 · question page 1650

AI.MEDIA.02 · Review generated subtitles and dubs in practice

A school posts a video with generated Spanish subtitles for families. What is the best step before posting?

Answer: B. Have a Spanish speaker review them

Review important translations.

- ✗ A. Shorten the video: irrelevant
- ✗ C. Post right away, since every family will understand the subtitles even

with small errors: may mislead

✗ D. Remove the English audio: unnecessary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F1-Q2 · question page 1651

AI.MEDIA.02 · Review generated subtitles and dubs in practice

Subtitles flash by too fast to read. Which check failed?

Answer: C. Timing

Lines must stay readable.

✗ A. None: timing failed

✗ B. Names: not names

✗ D. Meaning: separate

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F1-Q3 · question page 1652

AI.MEDIA.02 · Review generated subtitles and dubs in practice

A dub translates "Mrs. Kowalski" as "Mrs. Cow." What failed?

Answer: D. Names

Names must be right.

✗ A. Timing, since the dubbed name was said slightly later than in the original video: not stated

✗ B. Only the volume: not the issue

✗ C. Nothing: it failed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F1-Q4 · question page 1653

AI.MEDIA.02 · Review generated subtitles and dubs in practice

A subtitle says "5 miles" but the translated audio says "5 kilometers." What should you do?

Answer: B. Check which is right and fix the other

Units must match and be right.

- ✗ A. Delete the numbers: loses information
- ✗ C. Use the louder one: not a method
- ✗ D. Leave both, since miles and kilometers are about the same distance for most purposes: they differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F1-Q5 · question page 1654

AI.MEDIA.02 · Review generated subtitles and dubs in practice

Which check is most important for a safety video?

Answer: C. Meaning of each instruction

Wrong meaning can cause harm.

- ✗ A. The video length: less important
- ✗ B. The music volume: less important
- ✗ D. The font of the subtitles: less important

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F2-Q1 · question page 1655

AI.MEDIA.02 · Handle tone, consent and disclosure in translated media

A polite request in the original became a rude command in the dub. What changed?

Answer: A. The tone

Translation can shift politeness.

✗ B. The speaker's name: not names

✗ C. Nothing important: tone matters

✗ D. The timing: not timing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F2-Q2 · question page 1656

AI.MEDIA.02 · Handle tone, consent and disclosure in translated media

A dub uses a voice that imitates the original speaker. What question does this raise?

Answer: D. Whether the speaker consented

Imitated voices need consent.

✗ A. Whether the video has color: irrelevant

✗ B. Whether subtitles are needed: separate

✗ C. Whether the dub is long enough: not the issue

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F2-Q3 · question page 1657

AI.MEDIA.02 · Handle tone, consent and disclosure in translated media

A team publishes machine subtitles that no one fully reviewed. What should they add?

Answer: C. A note that subtitles were machine-generated

Disclose unreviewed translation.

✗ A. Nothing, since viewers always assume that every subtitle they read has been reviewed: not true

✗ B. More subtitles: not disclosure

✗ D. A longer title: not disclosure

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F2-Q4 · question page 1658

AI.MEDIA.02 · Handle tone, consent and disclosure in translated media

Why give viewers a way to report errors?

Answer: A. So mistakes can be found and fixed

Feedback improves accuracy.

- ✗ B. Because it is required for every video: not the reason
- ✗ C. To collect email addresses: not the goal
- ✗ D. So viewers can rewrite the entire video: not the goal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S79-F2-Q5 · question page 1659

AI.MEDIA.02 · Handle tone, consent and disclosure in translated media

A joke in the original falls flat in translation. What is the best approach?

Answer: B. Have a reviewer adapt it

Adapt meaning, not just words.

- ✗ A. Remove the whole scene: too drastic
- ✗ C. Add laughter sounds: not a fix
- ✗ D. Translate it word for word anyway, since the exact words matter more than the joke: loses meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S80-M-Q1 · question page 1661

AI.MEDIA.03 · Explain voice cloning and why consent and disclosure matter

What is voice cloning?

Answer: C. Making a synthetic voice that sounds like a real person

Synthetic speech in someone's voice.

- ✗ A. Recording someone's voice once and playing back exactly the same recording many times: that is replay
- ✗ B. Making music from noise: no
- ✗ D. Turning speech into text: that is transcription

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-M-Q2 · question page 1662

AI.MEDIA.03 · Explain voice cloning and why consent and disclosure matter

Which is a helpful use of voice cloning?

Answer: A. Helping someone who is losing their voice

Assistive use with consent.

- ✗ B. Faking a politician's statement: deceptive
- ✗ C. Making a fake celebrity endorsement: deceptive
- ✗ D. Calling a grandparent: a scam

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-M-Q3 · question page 1663

AI.MEDIA.03 · Explain voice cloning and why consent and disclosure matter

What does disclosure mean here?

Answer: B. Telling listeners the voice is synthetic

Being open about it.

- ✗ A. Keeping secret that the voice was made by a computer so the audio sounds more natural: opposite
- ✗ C. Making the audio louder: no
- ✗ D. Publishing the person's address: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-M-Q4 · question page 1664

AI.MEDIA.03 · Explain voice cloning and why consent and disclosure matter

What did Tennessee's ELVIS Act (effective July 1, 2024) do?

Answer: D. Added voice to that state's publicity rights

A state law on voice rights.

- ✗ A. Required all songs to be written by humans: not what it did
- ✗ B. Made voice cloning legal everywhere: not what it did
- ✗ C. Banned every use of synthetic voices anywhere: too broad

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-M-Q5 · question page 1665

AI.MEDIA.03 · Explain voice cloning and why consent and disclosure matter

What did the FCC rule in February 2024?

Answer: B. AI-generated voices in calls need prior consent

Treated as "artificial" voices.

- ✗ A. Phone calls must be recorded: not the ruling
- ✗ C. Voice cloning is a crime everywhere: not the ruling
- ✗ D. AI voices may be used in any phone call to anyone: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-B1-Q1 · question page 1666

AI.MEDIA.03 · Compare voice cloning with borrowing someone's signature

Why does a signature matter?

Answer: D. People believe you agreed when they see it

It stands for you.

- ✗ A. Because it is written in ink: not the point
- ✗ B. Because signatures are always the most beautiful part of any written document: not the reason
- ✗ C. Because it shows your height: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-B1-Q2 · question page 1667

AI.MEDIA.03 · Compare voice cloning with borrowing someone's signature

Copying a signature without permission is like what?

Answer: C. Cloning a voice without consent

Unauthorized use.

- ✗ A. Writing your own name: no
- ✗ B. Typing a letter: no
- ✗ D. Getting written permission: that is fine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-B1-Q3 · question page 1668

AI.MEDIA.03 · Compare voice cloning with borrowing someone's signature

Using a cloned voice with consent and disclosure is like what?

Answer: A. Signing with permission and a note

Permitted and open.

- ✗ B. Erasing a signature: no
- ✗ C. Losing a pen: no
- ✗ D. Forging a signature perfectly: the opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-B1-Q4 · question page 1669

AI.MEDIA.03 · Compare voice cloning with borrowing someone's signature

Why do listeners trust a familiar voice?

Answer: B. They believe the person is speaking

Voice signals identity.

- ✗ A. Because it is recorded: not the reason
- ✗ C. Because it is loud: no
- ✗ D. Because voices always tell the truth: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-B1-Q5 · question page 1670

AI.MEDIA.03 · Compare voice cloning with borrowing someone's signature

Is a perfect copy acceptable without permission?

Answer: A. No; permission is what matters

Consent, not quality.

- ✗ B. Yes, if it is short: no
- ✗ C. Yes, because a copy that is perfect cannot cause any harm to the person whose voice it is: it can
- ✗ D. Only on weekends: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-F1-Q1 · question page 1671

AI.MEDIA.03 · Apply consent and disclosure to voice cloning

A podcaster wants to use a cloned voice of a guest to fix a few words. What should come first?

Answer: C. The guest's clear permission

Consent before cloning.

- ✗ A. A louder microphone: irrelevant
- ✗ B. A longer episode: irrelevant
- ✗ D. Nothing, since small fixes to a few words never need anyone's permission at all: still their voice

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-F1-Q2 · question page 1672

AI.MEDIA.03 · Apply consent and disclosure to voice cloning

An ad uses a synthetic voice that sounds like a famous singer without permission. What is the concern?

Answer: D. It uses the singer's identity without consent

Unauthorized use of identity.

- ✗ A. The ad is too short: irrelevant
- ✗ B. The ad has music: irrelevant
- ✗ C. None, since a synthetic voice is a new voice: it imitates someone

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-F1-Q3 · question page 1673

AI.MEDIA.03 · Apply consent and disclosure to voice cloning

A school announcement uses a synthetic voice reading the principal's message, with permission. What should it add?

Answer: C. A note that the voice is synthetic

Disclosure.

- ✗ A. Nothing, since permission from the principal means listeners never need

to be told anything: disclosure still helps

✗ B. A different message: not needed

✗ D. Louder music: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-F1-Q4 · question page 1674

AI.MEDIA.03 · Apply consent and disclosure to voice cloning

Grandpa gets a call in his grandson's voice asking for money urgently. What should he do?

Answer: B. Hang up and call his grandson on a known number

Verify through a known channel.

✗ A. Give his bank details: unsafe

✗ C. Send the money right away, since the voice sounded exactly like his grandson: voices can be cloned

✗ D. Call back the number that called: may be the scammer

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-F1-Q5 · question page 1675

AI.MEDIA.03 · Apply consent and disclosure to voice cloning

A company sends robocalls with an AI voice and no prior consent. Under the FCC's 2024 ruling, what applies?

Answer: D. The same consent rules as other artificial-voice calls

Consent needed.

✗ A. A ban on all phone calls: too broad

✗ B. Only a fee: not stated

✗ C. No rules, because AI voices were left out of phone-call rules entirely by the ruling: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: law_elvis; law_fcc_voice

S80-F2-Q1 · question page 1676

AI.MEDIA.03 · Navigate differing laws and protect against voice scams

A voice use is allowed in one state. Is it allowed everywhere?

Answer: A. Not necessarily; laws differ by place

Check local law.

- ✗ B. Yes, if it is online: no
- ✗ C. No, it is banned everywhere: too strong
- ✗ D. Yes, because every state and every country follows exactly the same rules about voices: they differ

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-F2-Q2 · question page 1677

AI.MEDIA.03 · Navigate differing laws and protect against voice scams

What is a practical rule everywhere?

Answer: B. Get clear permission before cloning a voice

Permission first.

- ✗ A. Only clone famous people: worse
- ✗ C. Never tell anyone a voice is synthetic: opposite
- ✗ D. Clone any voice you like: still needs consent

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-F2-Q3 · question page 1678

AI.MEDIA.03 · Navigate differing laws and protect against voice scams

How can a family protect against fake voice calls?

Answer: C. Agree on a code word to verify each other

A shared secret.

- ✗ A. Post their voices online: risky
- ✗ B. Answer every call quickly and do whatever the familiar voice asks without any delay: risky
- ✗ D. Stop using phones: impractical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-F2-Q4 · question page 1679

AI.MEDIA.03 · Navigate differing laws and protect against voice scams

Are all legal questions about voice cloning settled?

Answer: A. No; courts are still deciding many

An evolving area.

- ✗ B. There are no laws about voices: some exist
- ✗ C. Yes, by the ELVIS Act for the whole world: one state
- ✗ D. Yes; every possible question about synthetic voices was answered by courts years ago: still developing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S80-F2-Q5 · question page 1680

AI.MEDIA.03 · Navigate differing laws and protect against voice scams

Which use most clearly crosses the line?

Answer: D. Imitating a real person to deceive listeners

Deception is the problem.

- ✗ A. A clearly labeled synthetic narrator: acceptable
- ✗ B. A screen reader voice: acceptable

✗ C. Using a synthetic voice, with consent: acceptable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_elvis; law_fcc_voice

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S81-M-Q1 · question page 1682

AI.MEDIA.04 · Check the rights and terms that apply before using generated music

What should you check first in a music tool's terms?

Answer: C. Whether commercial use is allowed

Use rights.

✗ A. The tool's logo: irrelevant
✗ B. The number of users: irrelevant
✗ D. The color scheme of the app: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-M-Q2 · question page 1683

AI.MEDIA.04 · Check the rights and terms that apply before using generated music

Per the US Copyright Office's January 2025 report, do prompts alone make you the author?

Answer: B. No

Prompts alone are not enough.

✗ A. Yes, because writing a detailed prompt always makes you the full author of every output: not per the report
✗ C. Only for songs under 3 minutes: no
✗ D. Only for long prompts: not per the report

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-M-Q3 · question page 1684

AI.MEDIA.04 · Check the rights and terms that apply before using generated music

Which can be protected by copyright, per that report?

Answer: A. Your own lyrics or meaningful edits

Human-authored contributions.

- ✗ B. Any random sound: no
- ✗ C. Someone else's melody: not yours
- ✗ D. The tool's output, in full, simply: not per the report

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-M-Q4 · question page 1685

AI.MEDIA.04 · Check the rights and terms that apply before using generated music

A generated track sounds almost exactly like a famous song. What should you do?

Answer: D. Avoid using it

Similarity risk.

- ✗ A. Speed it up slightly: not enough
- ✗ B. Change the title: not enough
- ✗ C. Use it: they can be similar

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-M-Q5 · question page 1686

AI.MEDIA.04 · Check the rights and terms that apply before using generated music

What record should you keep?

Answer: B. The tool, plan and date

Proof of terms at the time.

- ✗ A. Nothing: helps to keep records
- ✗ C. The tool's home page design: irrelevant

✗ D. Only the song's title: may change

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-B1-Q1 · question page 1687

AI.MEDIA.04 · Compare music rights with borrowing a library book

What rules come with a library book?

Answer: A. You can read it but not sell copies

Use limits.

✗ B. You must memorize it: no
✗ C. You must read it in one day: not the rule
✗ D. You can do anything at all with it: not allowed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-B1-Q2 · question page 1688

AI.MEDIA.04 · Compare music rights with borrowing a library book

In music tools, what sets the rules?

Answer: D. The tool's terms

Terms decide use.

✗ A. The song's length: not the source of rules
✗ B. The weather: no
✗ C. The listener's mood: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-B1-Q3 · question page 1689

AI.MEDIA.04 · Compare music rights with borrowing a library book

Writing your name in a library book is like what?

Answer: C. Typing a prompt

Not authorship by itself.

- ✗ A. Composing the music yourself from scratch, note by note, without any tool at all: that is authorship
- ✗ B. Buying the book: no
- ✗ D. Returning the book: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-B1-Q4 · question page 1690

AI.MEDIA.04 · Compare music rights with borrowing a library book

A free plan allowing only personal use is like what?

Answer: D. Reading a library book at home

Limited use.

- ✗ A. Owning the library: no
- ✗ B. Writing a new book: no
- ✗ C. Photocopying the library book: not allowed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-B1-Q5 · question page 1691

AI.MEDIA.04 · Compare music rights with borrowing a library book

What is the lesson?

Answer: A. Check the rules before using generated music

Read the terms.

- ✗ B. Generated music has no rules at all, so it can be used anywhere for

anything: it has terms

✗ C. Never listen to music: no

✗ D. Only use library music: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F1-Q1 · question page 1692

AI.MEDIA.04 · Use generated music responsibly in practice

Eli wants generated background music for a monetized video. What should he check?

Answer: C. That his plan allows commercial use

Commercial use rights.

✗ A. Nothing, because all generated music may be used in any video, including monetized ones: depends on terms

✗ B. Only the song length: not enough

✗ D. Only the tempo: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F1-Q2 · question page 1693

AI.MEDIA.04 · Use generated music responsibly in practice

A student writes her own lyrics and uses a tool for the beat. Which part is most clearly hers?

Answer: B. The lyrics

Human-authored contribution.

✗ A. The tool's code: no

✗ C. Neither: the lyrics are hers

✗ D. The generated beat: prompts alone are not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F1-Q3 · question page 1694

AI.MEDIA.04 · Use generated music responsibly in practice

A prompt asks for a song "in the voice of" a named living singer. What is the concern?

Answer: D. Imitating a real artist

Avoid imitating real artists.

- ✗ A. The song may be too slow: irrelevant
- ✗ B. The song may be too short: irrelevant
- ✗ C. None: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F1-Q4 · question page 1695

AI.MEDIA.04 · Use generated music responsibly in practice

A tool's terms say "credit required." What should you do?

Answer: C. Include the credit as described

Follow the terms.

- ✗ A. Credit yourself only: not what is required
- ✗ B. Change tools after publishing: too late
- ✗ D. Leave out the credit: required

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F1-Q5 · question page 1696

AI.MEDIA.04 · Use generated music responsibly in practice

Why save proof of your plan when you generated a track?

Answer: B. Terms can change later

Records protect you.

✗ A. Because it is required by every country: not the reason

✗ C. Because tools delete every song that was made on a free plan after one week: not the reason

✗ D. To make the song louder: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F2-Q1 · question page 1697

AI.MEDIA.04 · Reduce rights risks while the law develops

Are rights questions about generated music fully settled?

Answer: A. No; they are still being argued

An open area.

✗ B. There are no rules: there are

✗ C. Yes, the same in every country: they differ

✗ D. Yes; one court case: not settled

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F2-Q2 · question page 1698

AI.MEDIA.04 · Reduce rights risks while the law develops

Which terms should you rely on?

Answer: C. The current terms

Read the current version.

✗ A. Any terms you like: no

✗ B. A friend's memory of the terms: unreliable

✗ D. Terms from a blog post written several years ago about a different version of the tool: may be outdated

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F2-Q3 · question page 1699

AI.MEDIA.04 · Reduce rights risks while the law develops

A platform flags your video because its generated music matches an existing recording. What does this show?

Answer: B. Generated music can resemble existing works

Similarity can be detected.

- ✗ A. That videos need no music: no
- ✗ C. That the platform is always wrong, since generated songs cannot match existing recordings: they can
- ✗ D. That music is banned: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F2-Q4 · question page 1700

AI.MEDIA.04 · Reduce rights risks while the law develops

For a school concert, what should guide the music choice?

Answer: D. The venue's rules on music

Follow the venue's rules.

- ✗ A. The loudest song: not a rule
- ✗ B. Whatever is trending: not a rule
- ✗ C. Only what the tool allows: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S81-F2-Q5 · question page 1701

AI.MEDIA.04 · Reduce rights risks while the law develops

What does this course do on disputed music rights?

Answer: A. It does not settle them

Disputed questions stay open.

- ✗ B. It says all generated music is illegal: not the position
- ✗ C. It gives one final answer that applies in every country for every kind of song: it does not
- ✗ D. It says anything goes: not the position

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_usco_part2

S82-M-Q1 · question page 1703

AI.MEDIA.05 · Check video for consistency over time and disclose synthetic presenters

What challenge does video add compared with single images?

Answer: A. Consistency over time

Frames must stay consistent.

- ✗ B. Videos have no people: no
- ✗ C. Videos have no colors: no
- ✗ D. Video can never be generated by AI tools, so it is always real footage from a camera: it can be

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-M-Q2 · question page 1704

AI.MEDIA.05 · Check video for consistency over time and disclose synthetic presenters

Which is a sign of a generated video?

Answer: C. A shirt that changes pattern between frames

Inconsistency over time.

- ✗ A. A video with music: not a sign
- ✗ B. A short video: not a sign

✗ D. A video that was recorded in a bright room: not a sign

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-M-Q3 · question page 1705

AI.MEDIA.05 · Check video for consistency over time and disclose synthetic presenters

What should you check about lip movement?

Answer: B. Whether it matches the words

Lip sync.

✗ A. Whether the person has a microphone: not the check
✗ C. Whether the person is smiling: not the check
✗ D. Whether the lips are a natural color: not the main check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-M-Q4 · question page 1706

AI.MEDIA.05 · Check video for consistency over time and disclose synthetic presenters

What is a synthetic presenter?

Answer: D. An AI-generated person or avatar presenting a video

A generated presenter.

✗ A. A microphone: no
✗ B. A video file format: no
✗ C. A real news anchor who reads the news from a teleprompter: real person

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-M-Q5 · question page 1707

AI.MEDIA.05 · Check video for consistency over time and disclose synthetic presenters

How should a synthetic presenter be disclosed?

Answer: A. With an on-screen label or description note

Tell viewers clearly.

- ✗ B. It does not need disclosing if it looks realistic enough that viewers will not notice: opposite
- ✗ C. By adding music: not disclosure
- ✗ D. By making the video shorter: not disclosure

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-B1-Q1 · question page 1708

AI.MEDIA.05 · Compare video consistency with continuity in movies

What is the continuity job in movies?

Answer: C. Keeping details the same between shots

Consistent details.

- ✗ A. Selling tickets: no
- ✗ B. Choosing actors: no
- ✗ D. Writing the music that plays during the final scenes of the movie for the audience: not continuity

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-B1-Q2 · question page 1709

AI.MEDIA.05 · Compare video consistency with continuity in movies

A coffee cup half full, then full in the next shot, is what?

Answer: D. A continuity slip

Details changed.

✗ A. Good editing: it is a slip

✗ B. A lighting choice: no

✗ C. A clever special effect: not intended

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-B1-Q3 · question page 1710

AI.MEDIA.05 · Compare video consistency with continuity in movies

A necklace disappearing in a generated video is like what?

Answer: B. A continuity slip

Inconsistency.

✗ A. A sound error: no

✗ C. A subtitle error: no

✗ D. Perfect continuity: not realistic

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-B1-Q4 · question page 1711

AI.MEDIA.05 · Compare video consistency with continuity in movies

Who notices continuity slips?

Answer: A. Viewers

They are visible.

✗ B. Only the director: viewers too

✗ C. Only the camera: no

✗ D. Nobody: they notice

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-B1-Q5 · question page 1712

AI.MEDIA.05 · Compare video consistency with continuity in movies

What should you look for in generated video?

Answer: B. Details changing between moments

Continuity checks.

- ✗ A. Only the length: not enough
- ✗ C. Only the music: not enough
- ✗ D. Only the title at the start: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F1-Q1 · question page 1713

AI.MEDIA.05 · Check and disclose video in practice

A training video uses an AI avatar to present safety rules. What should it include?

Answer: C. A clear note that the presenter is AI-generated

Disclose synthetic presenters.

- ✗ A. Nothing extra, since viewers only care about the safety rules and not about who presents them: disclosure matters
- ✗ B. A fake name and job title: deceptive
- ✗ D. A claim the presenter is real: deceptive

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F1-Q2 · question page 1714

AI.MEDIA.05 · Check and disclose video in practice

In a clip, a person's earrings vanish and return. What does this suggest?

Answer: D. Possible generation or editing

Inconsistency is a clue.

✗ A. A bad microphone: irrelevant

✗ B. A slow internet connection: not likely

✗ C. Nothing: not normal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F1-Q3 · question page 1715

AI.MEDIA.05 · Check and disclose video in practice

A video's words do not match the speaker's lip movements. What should you consider?

Answer: C. The audio may have been changed or generated

Lip sync mismatch.

✗ A. Nothing: worth checking

✗ B. That the video is very old: not necessarily

✗ D. That the speaker has a cold, which always makes lips move differently from the words: not likely

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F1-Q4 · question page 1716

AI.MEDIA.05 · Check and disclose video in practice

A ball passes through a wall in a video meant to look real. What does this break?

Answer: A. Ordinary physics

Physics errors.

✗ B. Lip sync, since the ball and the wall are both moving their lips in the scene: no

✗ C. Subtitles: no

✗ D. Nothing: it is impossible

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F1-Q5 · question page 1717

AI.MEDIA.05 · Check and disclose video in practice

A small business makes an ad with a synthetic spokesperson. What is required by honesty?

Answer: D. Telling viewers the spokesperson is synthetic

Disclosure.

- ✗ A. Using more music: irrelevant
- ✗ B. Hiding the business name: irrelevant
- ✗ C. Saying the spokesperson is a real customer: deceptive

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F2-Q1 · question page 1718

AI.MEDIA.05 · Handle subtle edits and context in video

A real video has one word changed by AI. Why can this matter?

Answer: B. It can change the meaning

Small edits, big meaning.

- ✗ A. It changes the colors: irrelevant
- ✗ C. It makes the file bigger: irrelevant
- ✗ D. It cannot matter: it can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F2-Q2 · question page 1719

AI.MEDIA.05 · Handle subtle edits and context in video

A clip is cut so a person seems to say the opposite of what they meant. What is the problem?

Answer: A. Removed context misleads

Context matters.

✗ B. The clip is too long: not the issue

✗ C. The clip has no music: not the issue

✗ D. No problem: context matters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F2-Q3 · question page 1720

AI.MEDIA.05 · Handle subtle edits and context in video

When publishing an edited video, what should you keep?

Answer: B. The original footage

Originals show what changed.

✗ A. Nothing: keep originals

✗ C. Only the audio: not enough

✗ D. Only the edited version: keep it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F2-Q4 · question page 1721

AI.MEDIA.05 · Handle subtle edits and context in video

A synthetic presenter looks like a real local teacher. What is needed?

Answer: D. The teacher's consent

Likeness needs consent.

✗ A. A different background: irrelevant

✗ B. A longer video: irrelevant

✗ C. Nothing: consent needed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S82-F2-Q5 · question page 1722

AI.MEDIA.05 · Handle subtle edits and context in video

Which use should never happen?

Answer: C. Putting words in a real person's mouth

Fake statements by real people.

- ✗ A. Replacing a plain background with a nicer one and noting the change in the video description: acceptable
- ✗ B. Adding captions: acceptable
- ✗ D. Trimming silence: acceptable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S83-M-Q1 · question page 1724

AI.MEDIA.06 · Verify surprising media by checking source and context

What is a deepfake?

Answer: D. Realistic fake media showing someone doing what they did not

Realistic and false.

- ✗ A. A cartoon: no
- ✗ B. A very old photo: no
- ✗ C. Any video that has been edited at all, including trimming silence: too broad

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-M-Q2 · question page 1725

AI.MEDIA.06 · Verify surprising media by checking source and context

Why is looking for one visual oddity unreliable?

Answer: B. Real videos have glitches and new fakes may have none

Looks can mislead both ways.

✗ A. Because oddities are always invisible: no

✗ C. Because all videos are fake: no

✗ D. Because fakes always show strange blinking, so if blinking looks normal the video is real: not reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-M-Q3 · question page 1726

AI.MEDIA.06 · Verify surprising media by checking source and context

Which is a source check?

Answer: C. Finding who first posted it

Origin matters.

✗ A. Looking at the thumbnail: not source

✗ B. Counting how many times the person in the video blinks during the first ten seconds: visual, not source

✗ D. Checking the video length: not source

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-M-Q4 · question page 1727

AI.MEDIA.06 · Verify surprising media by checking source and context

What can a reverse image search find?

Answer: A. Older, original versions of an image

Earlier versions.

✗ B. The video's future views: no

✗ C. The camera's battery level: no

✗ D. The exact person who made a fake, along with their home address and phone number: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: law_takeitdown

S83-M-Q5 · question page 1728

AI.MEDIA.06 · Verify surprising media by checking source and context

Which clip deserves the most checking?

Answer: B. A surprising one that makes you angry or scared

Strong emotions, more checks.

- ✗ A. A weather report: low risk
- ✗ C. A calm, ordinary clip of a local park on a sunny afternoon with nothing much happening: low risk
- ✗ D. A cooking video: low risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-B1-Q1 · question page 1729

AI.MEDIA.06 · Verify a clip with four questions

A clip posted by a brand-new anonymous account fails which question?

Answer: C. Source

Untrustworthy origin.

- ✗ A. Earlier versions: not shown
- ✗ B. Confirmation: not shown
- ✗ D. Context: it is source

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-B1-Q2 · question page 1730

AI.MEDIA.06 · Verify a clip with four questions

Snow on a 35 °C day fails which question?

Answer: A. Context

Facts do not fit.

- ✗ B. Confirmation: not shown
- ✗ C. Earlier versions: not shown
- ✗ D. Source: it is context

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-B1-Q3 · question page 1731

AI.MEDIA.06 · Verify a clip with four questions

No trusted outlet reports a "major" event. Which question fails?

Answer: D. Confirmation

No confirmation.

- ✗ A. Context: separate
- ✗ B. Source: separate
- ✗ C. Earlier versions: not shown

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-B1-Q4 · question page 1732

AI.MEDIA.06 · Verify a clip with four questions

A reverse search finds the same photo from 2018 with a different caption. Which question?

Answer: B. Earlier versions

Recycled media.

- ✗ A. Context: separate
- ✗ C. Source: separate
- ✗ D. Confirmation: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-B1-Q5 · question page 1733

AI.MEDIA.06 · Verify a clip with four questions

How many of the four questions does the snow example fail?

Answer: D. Two

Source and context.

- ✗ A. One: two
- ✗ B. None: two
- ✗ C. Four: only two shown

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-F1-Q1 · question page 1734

AI.MEDIA.06 · Verify surprising media in practice

A video shows a mayor announcing schools will close tomorrow. What should you check first?

Answer: C. The city's official accounts or trusted news

Official confirmation.

- ✗ A. The video's music: irrelevant
- ✗ B. How many likes it has: popularity is not proof
- ✗ D. Whether the mayor blinks normally in the video: unreliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-F1-Q2 · question page 1735

AI.MEDIA.06 · Verify surprising media in practice

A dramatic photo of a flood is going viral. A reverse search finds it from another country years ago. What does this show?

Answer: A. It is being shared with false context

Real image, false context.

- ✗ B. Nothing: it shows misuse
- ✗ C. That the photo is AI-generated: it is real but recycled
- ✗ D. That the flood is happening again: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-F1-Q3 · question page 1736

AI.MEDIA.06 · Verify surprising media in practice

A friend shares a shocking audio clip of a celebrity. The clip has no source. What is the best response?

Answer: B. Do not share it until it is confirmed

Verify before sharing.

- ✗ A. Report the friend to police: excessive
- ✗ C. Add your own comments and repost: spreads it
- ✗ D. Share it with a note saying "might be fake,": still spreads

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-F1-Q4 · question page 1737

AI.MEDIA.06 · Verify surprising media in practice

Which detail is the strongest sign a clip is genuine?

Answer: D. Several trusted outlets confirm it independently

Independent confirmation.

- ✗ A. It was posted recently: not proof
- ✗ B. It has many views: not proof
- ✗ C. The clip looks very high quality and sharp: quality is not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-F1-Q5 · question page 1738

AI.MEDIA.06 · Verify surprising media in practice

A clip shows a protest "today," but trees are bare and it is summer. Which check does this use?

Answer: C. Context

Season does not fit.

- ✗ A. Earlier versions: not this
- ✗ B. Confirmation: not this
- ✗ D. Source: not source

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-F2-Q1 · question page 1739

AI.MEDIA.06 · Avoid both mistakes and respond to harmful fakes

What are the two common mistakes?

Answer: A. Believing fakes and dismissing real evidence as AI

Both errors.

- ✗ B. Using reverse search and reading captions: good habits
- ✗ C. Watching videos and reading articles: not mistakes
- ✗ D. Checking sources too carefully and asking trusted news outlets whether events really happened: good habits

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-F2-Q2 · question page 1740

AI.MEDIA.06 · Avoid both mistakes and respond to harmful fakes

Someone says a real recording of them is "probably AI" to avoid blame. What is this an example

of?

Answer: B. Dismissing real evidence

Denying real media.

- ✗ A. Disclosure: no
- ✗ C. Good fact-checking: not checking
- ✗ D. Believing a fake: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-F2-Q3 · question page 1741

AI.MEDIA.06 · Avoid both mistakes and respond to harmful fakes

What avoids both mistakes?

Answer: A. Checking sources rather than trusting looks

Source-based checking.

- ✗ B. Trusting your first impression of how real a video looks, without any further checks: causes both
- ✗ C. Never watching videos: impractical
- ✗ D. Believing whatever is most popular: unreliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-F2-Q4 · question page 1742

AI.MEDIA.06 · Avoid both mistakes and respond to harmful fakes

From May 19, 2026, what must covered US platforms do under the TAKE IT DOWN Act?

Answer: C. Offer a removal process and act on valid requests within 48 hours

A dated federal requirement.

- ✗ A. Nothing new: it adds duties

✗ B. Ban all video uploads: no

✗ D. Remove every AI-generated image from their sites, including art: too broad

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S83-F2-Q5 · question page 1743

AI.MEDIA.06 · Avoid both mistakes and respond to harmful fakes

You receive a nonconsensual intimate deepfake of a classmate. What should you do?

Answer: D. Report it and do not share it

Report, do not spread.

✗ A. Keep it to show later: harmful

✗ B. Post it with a warning: spreads harm

✗ C. Forward it to a few friends: spreads harm

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: law_takeitdown

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S84-M-Q1 · question page 1745

AI.MEDIA.07 · Explain provenance labels and the limits of detection

What does provenance mean?

Answer: A. A record of where content came from and how it changed

Origin and history.

✗ B. The number of views: no

✗ C. A rating of how beautiful or well made a picture is compared with other pictures online: not the meaning

✗ D. The file size: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-M-Q2 · question page 1746

AI.MEDIA.07 · Explain provenance labels and the limits of detection

What are Content Credentials based on?

Answer: C. The C2PA standard

Coalition for Content Provenance and Authenticity.

- ✗ A. A social media app: no
- ✗ B. A camera brand: no
- ✗ D. A single government law: it is a standard

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-M-Q3 · question page 1747

AI.MEDIA.07 · Explain provenance labels and the limits of detection

How can provenance records be lost?

Answer: B. By screenshots or re-uploading

Records can be stripped.

- ✗ A. By looking at it: no
- ✗ C. By opening the file: not usually
- ✗ D. They can never be lost once they are attached, no matter what happens to the file: they can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-M-Q4 · question page 1748

AI.MEDIA.07 · Explain provenance labels and the limits of detection

An image has no provenance label. What does that prove?

Answer: D. Nothing about whether it is real or fake

Missing labels are not proof.

- ✗ A. That it is real: not proven
- ✗ B. That it is old: not proven
- ✗ C. That the image must be fake, because every real photo always carries a provenance label: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

---

S84-M-Q5 · question page 1749

AI.MEDIA.07 · Explain provenance labels and the limits of detection

What do detector tools give?

Answer: B. Probabilities that can be wrong both ways

Probabilistic and fallible.

- ✗ A. A copyright license: no
- ✗ C. The creator's name: no
- ✗ D. Certain answers that prove beyond doubt: not certain

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

---

S84-B1-Q1 · question page 1750

AI.MEDIA.07 · Compare provenance with a package tracking history

What does a tracking history show?

Answer: D. Where the parcel has been

Its history.

- ✗ A. The driver's name only: not the main point
- ✗ B. The weather: no
- ✗ C. What is inside the parcel: not the purpose

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: media_c2pa

S84-B1-Q2 · question page 1751

AI.MEDIA.07 · Compare provenance with a package tracking history

Provenance for a file is like what?

Answer: A. A tracking history

Origin and changes.

- ✗ B. A gift card: no
- ✗ C. A price tag: no
- ✗ D. A brand-new plain box: that is missing provenance

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-B1-Q3 · question page 1752

AI.MEDIA.07 · Compare provenance with a package tracking history

Repacking the item in a plain box is like what?

Answer: C. Stripping provenance by re-uploading

Records lost.

- ✗ A. Delivering faster: no
- ✗ B. Opening the parcel: no
- ✗ D. Adding a new and more detailed tracking label that shows the parcel's full history: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-B1-Q4 · question page 1753

AI.MEDIA.07 · Compare provenance with a package tracking history

A box with no tracking proves what?

Answer: B. Not necessarily anything

Absence is not proof.

- ✗ A. That it is valuable: no
- ✗ C. That the box was definitely stolen and should be reported to the police at once: not proof
- ✗ D. That it is empty: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-B1-Q5 · question page 1754

AI.MEDIA.07 · Compare provenance with a package tracking history

What is the lesson?

Answer: C. Provenance helps but its absence proves little

Useful but limited.

- ✗ A. Only buy online: no
- ✗ B. Tracking is useless: too strong
- ✗ D. Never trust anything that has a tracking history: not the lesson

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-F1-Q1 · question page 1755

AI.MEDIA.07 · Use provenance and detection carefully in practice

A news photo shows Content Credentials listing the camera and edits. What does this give you?

Answer: A. Useful information about its history

A helpful clue.

- ✗ B. A copyright license: no
- ✗ C. Nothing useful: it helps
- ✗ D. Absolute proof that everything shown in the photo is true, so no other

checks are needed: still check context

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-F1-Q2 · question page 1756

AI.MEDIA.07 · Use provenance and detection carefully in practice

A viral image has no credentials because it was screenshotted. What should you conclude?

Answer: D. Nothing yet; check other clues

Absence proves little.

- ✗ A. That it is fake, because only fake images lose their credentials when they are shared: not proof
- ✗ B. That it is old: not proven
- ✗ C. That it is real: not proven

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-F1-Q3 · question page 1757

AI.MEDIA.07 · Use provenance and detection carefully in practice

A detector says a student essay's photo is "70% likely AI." What should a teacher do?

Answer: B. Treat it as one clue and talk with the student

Scores are not proof.

- ✗ A. Ignore the photo: not necessary
- ✗ C. Report the student publicly: harmful
- ✗ D. Give the student a failing grade right away: not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-F1-Q4 · question page 1758

AI.MEDIA.07 · Use provenance and detection carefully in practice

A generator adds a visible watermark. What can remove it?

Answer: C. Cropping or editing

Watermarks can be removed.

- ✗ A. Viewing the image: no
- ✗ B. Nothing at all, because watermarks are permanent and cannot be removed by any method: they can be
- ✗ D. Saving the file: not usually

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-F1-Q5 · question page 1759

AI.MEDIA.07 · Use provenance and detection carefully in practice

Which approach to a suspicious image is best?

Answer: D. Combine provenance with source and context checks

Several clues together.

- ✗ A. Rely only on how it looks: unreliable
- ✗ B. Rely only on likes: unreliable
- ✗ C. Rely only on a detector score, since detectors are designed to replace every other check: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

---

S84-F2-Q1 · question page 1760

AI.MEDIA.07 · Avoid misusing detector scores

What is a false positive for a detector?

Answer: A. Flagging a real photo as AI

Wrongly flagged.

✗ B. Correctly passing a real photo: true negative

✗ C. Correctly flagging an AI image: true positive

✗ D. Missing an AI image: false negative

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-F2-Q2 · question page 1761

AI.MEDIA.07 · Avoid misusing detector scores

What is a false negative?

Answer: C. Missing an AI image

A fake that passes.

✗ A. Correctly passing a real photo: true negative

✗ B. Correctly finding an AI image: true positive

✗ D. Flagging a real photo as AI: false positive

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

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S84-F2-Q3 · question page 1762

AI.MEDIA.07 · Avoid misusing detector scores

Why is accusing someone based on a score alone harmful?

Answer: D. Scores can be wrong

Errors cause real harm.

✗ A. Because detector scores are always exactly 50%, so they never give any information: not true

✗ B. Because scores are secret: no

✗ C. Because accusations are always illegal: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F2-Q4 · question page 1763

AI.MEDIA.07 · Avoid misusing detector scores

A journalist's real photo is flagged "90% AI." What should an editor do?

Answer: B. Check the source and camera files

Investigate further.

- ✗ A. Retract the photo at once, since 90% is high enough to treat the score as certain proof: not certain
- ✗ C. Fire the journalist: harmful
- ✗ D. Publish the score instead: not helpful

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S84-F2-Q5 · question page 1764

AI.MEDIA.07 · Avoid misusing detector scores

How should provenance and detection be treated?

Answer: A. As one clue among several

One clue of many.

- ✗ B. As useless: they help
- ✗ C. As legal proof: not proof
- ✗ D. As the final answer: not reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: media_c2pa

S85-M-Q1 · question page 1766

AI.CODE.01 · Read a few lines of Python and predict what they do

In this example, what does the program print?

Prepared code example (Python 3). Shown only; nothing runs.

```
price = 4
count = 3
total = price * count
print(total)
```

Answer: C. 12

4 * 3 = 12.

- ✗ A. 7, because the program adds the price and the count together before printing the total: it multiplies
- ✗ B. total: it prints the value, not the name
- ✗ D. 4: that is price

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S85-M-Q2 · question page 1767

AI.CODE.01 · Read a few lines of Python and predict what they do

What does "price = 4" do in Python?

Answer: A. Stores 4 under the name price

"=" stores a value.

- ✗ B. Deletes price: no
- ✗ C. Checks whether price is already equal to 4 and prints True or False on the screen: that is ==
- ✗ D. Prints 4: no print

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-M-Q3 · question page 1768

AI.CODE.01 · Read a few lines of Python and predict what they do

If the second line were "count = 5," what would print?

Prepared code example (Python 3). Shown only; nothing runs.

```
price = 4
count = 3
total = price * count
print(total)
```

Answer: D. 20

4 * 5 = 20.

- ✗ A. 9, because changing the count to 5 adds 5 to the price of 4 instead of multiplying: it multiplies
- ✗ B. 5: that is count
- ✗ C. 12: count changed

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S85-M-Q4 · question page 1769

AI.CODE.01 · Read a few lines of Python and predict what they do

In what order does Python run these lines?

Answer: B. Top to bottom

Instructions run in order.

- ✗ A. In random order: no
- ✗ C. Only the last line: no
- ✗ D. Bottom to top: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-M-Q5 · question page 1770

AI.CODE.01 · Read a few lines of Python and predict what they do

What does print(...) do?

Answer: C. Shows a value on screen

Displays output.

- ✗ A. Multiplies numbers: that is *
- ✗ B. Stores a value: that is =
- ✗ D. Sends the program: not in Python

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-B1-Q1 · question page 1771

AI.CODE.01 · Trace variables on paper

After line 2, what is b?

Answer: A. 5

2 + 3 = 5.

- ✗ B. 3: a + 3
- ✗ C. 10: later
- ✗ D. 2, because b is given the same value as a, and the "+ 3" part is ignored by Python: adds 3

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S85-B1-Q2 · question page 1772

AI.CODE.01 · Trace variables on paper

After line 3, what is a?

Answer: B. 10

a is replaced.

- ✗ A. 12: no
- ✗ C. 2, because once a variable is given a value it can never be changed by a later line: it changes

✗ D. 5: that is b

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-B1-Q3 · question page 1773

AI.CODE.01 · Trace variables on paper

After line 3, what is b?

Answer: D. 5

b was already calculated.

✗ A. 10: that is a
✗ B. 2: no
✗ C. 13: it does not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-B1-Q4 · question page 1774

AI.CODE.01 · Trace variables on paper

What does line 4 print?

Answer: C. 15

10 + 5 = 15.

✗ A. 12: no
✗ B. 7, which is the original value of a added to the original value of b from line 2: a changed
✗ D. a + b: prints the value

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S85-B1-Q5 · question page 1775

AI.CODE.01 · Trace variables on paper

What is the key tracing habit?

Answer: D. Update a variable only when a line changes it

Cross out and rewrite.

- ✗ A. Track only the last line: no
- ✗ B. Update every variable on every line, even when the line does not mention that variable: no
- ✗ C. Ignore numbers: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F1-Q1 · question page 1776

AI.CODE.01 · Predict the output of short programs

What does this program print?

Prepared code example (Python 3). Shown only; nothing runs.

```
age = 15
if age >= 16:
    print("Can apply")
else:
    print("Too young")
```

Answer: B. Too young

15 >= 16 is false, so else runs.

- ✗ A. 15: not printed
- ✗ C. Nothing: else runs
- ✗ D. Can apply: condition false

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S85-F1-Q2 · question page 1777

AI.CODE.01 · Predict the output of short programs

If the first line were "age = 16," what would print?

Prepared code example (Python 3). Shown only; nothing runs.

```
age = 15
if age >= 16:
    print("Can apply")
else:
    print("Too young")
```

Answer: A. Can apply

16 >= 16 is true.

- ✗ B. 16: not printed
- ✗ C. Both lines: only one branch runs
- ✗ D. Too young: >= includes 16

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S85-F1-Q3 · question page 1778

AI.CODE.01 · Predict the output of short programs

What does this loop print?

Prepared code example (Python 3). Shown only; nothing runs.

```
for n in [1, 2, 3]:
    print(n * 2)
```

Answer: D. 2, 4 and 6 on separate lines

Each n doubled.

- ✗ A. 12: no sum
- ✗ B. 6 only: prints each time
- ✗ C. 1, 2 and 3 on separate lines: it doubles

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S85-F1-Q4 · question page 1779

AI.CODE.01 · Predict the output of short programs

How many times does the loop body run?

Prepared code example (Python 3). Shown only; nothing runs.

```
for n in [1, 2, 3]:
    print(n * 2)
```

Answer: B. 3

Three items in the list.

- ✗ A. 6, because the loop runs once for every number that it prints out on the screen: once per item
- ✗ C. 2: once per item
- ✗ D. 1: once per item

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S85-F1-Q5 · question page 1780

AI.CODE.01 · Predict the output of short programs

In "x = 7" then "x = x + 1," what is x afterward?

Answer: A. 8

7 + 1 = 8.

- ✗ B. 7, because a variable cannot be used on both sides of the same "=" sign in Python: it can
- ✗ C. 71: not text joining
- ✗ D. 1: no

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F2-Q1 · question page 1781

AI.CODE.01 · Read if statements, loops and indentation

What does "if age >= 16:" mean?

Answer: C. If age is at least 16

Greater than or equal.

- ✗ A. Set age to 16: not assignment
- ✗ B. If age is under 16: reversed
- ✗ D. If age is exactly 16: >= includes more

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F2-Q2 · question page 1782

AI.CODE.01 · Read if statements, loops and indentation

What does indentation show in Python?

Answer: A. Which lines belong together

Blocks of code.

- ✗ B. Comments: no
- ✗ C. How important each line is, with the most important lines indented the furthest: not importance
- ✗ D. Nothing; it is decoration: it matters

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F2-Q3 · question page 1783

AI.CODE.01 · Read if statements, loops and indentation

In "for n in [5, 10]: print(n)," what prints?

Answer: C. 5 then 10

Each item in turn.

- ✗ A. Nothing: it prints
- ✗ B. n twice: prints values
- ✗ D. 15: no sum

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F2-Q4 · question page 1784

AI.CODE.01 · Read if statements, loops and indentation

When does an else part run?

Answer: D. When the if condition is false

The other branch.

- ✗ A. Never: it can
- ✗ B. Always, right after the if part has finished, whether the condition was true or false: only when false
- ✗ C. Before the if: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S85-F2-Q5 · question page 1785

AI.CODE.01 · Read if statements, loops and indentation

A loop goes over a list of 4 names. How many times does its body run?

Answer: B. 4

Once per item.

- ✗ A. 8: no
- ✗ C. 0: it runs
- ✗ D. 1, because a loop body runs only once no matter how many items are in

the list: once per item

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-M-Q1 · question page 1787

AI.CODE.02 · Explain why code that looks right can be wrong and how to check it

Why can generated code be wrong even when it looks neat?

Answer: A. The model predicts likely code without proving it

Prediction, not proof.

- ✗ B. Because computers cannot run neat code: no
- ✗ C. Because comments stop code from running: no
- ✗ D. Because neat code is always wrong, and only messy code can actually work correctly: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-M-Q2 · question page 1788

AI.CODE.02 · Explain why code that looks right can be wrong and how to check it

In this example, which result is wrong?

Prepared code example (Python 3). Shown only; nothing runs.

```
def average(numbers):
    return sum(numbers) / 2

print(average([4, 6]))
print(average([3, 6, 9]))
```

Answer: D. 9.0 for [3, 6, 9]

The average of 3, 6, 9 is 6.

- ✗ A. Both results: 5.0 is right

✗ B. Neither result: one is wrong

✗ C. 5.0 for [4, 6]: 5.0 is right

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S86-M-Q3 · question page 1789

AI.CODE.02 · Explain why code that looks right can be wrong and how to check it

What is the bug in the same example?

Prepared code example (Python 3). Shown only; nothing runs.

```
def average(numbers):
    return sum(numbers) / 2

print(average([4, 6]))
print(average([3, 6, 9]))
```

Answer: C. It always divides by 2

It should divide by len(numbers).

✗ A. Its name is wrong: not the bug

✗ B. It prints too much: not the bug

✗ D. It uses sum: sum is fine

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S86-M-Q4 · question page 1790

AI.CODE.02 · Explain why code that looks right can be wrong and how to check it

What does the example show about testing?

Prepared code example (Python 3). Shown only; nothing runs.

```
def average(numbers):
    return sum(numbers) / 2

```

```
print(average([4, 6]))
print(average([3, 6, 9]))
```

Answer: B. Passing one test does not prove code right

Test several inputs.

✗ A. Code never needs tests: no

✗ C. Testing is a waste of time, because the first test already showed that the function works: reversed

✗ D. Two tests are always enough: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S86-M-Q5 · question page 1791

AI.CODE.02 · Explain why code that looks right can be wrong and how to check it

What is the best way to check code?

Answer: C. Run it on inputs whose answers you know

Known answers reveal errors.

✗ A. Count its lines: not a check

✗ B. Ask the tool that wrote it whether it is correct, and accept the answer that it gives: not independent

✗ D. Check that it has comments: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-B1-Q1 · question page 1792

AI.CODE.02 · Compare checking code with checking a calculator you built

Why type 2 + 2 into a new calculator?

Answer: B. You already know the answer

Known answers reveal errors.

- ✗ A. To make it faster: no
- ✗ C. To clean the buttons: no
- ✗ D. Because 2 + 2 is the only problem: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-B1-Q2 · question page 1793

AI.CODE.02 · Compare checking code with checking a calculator you built

The calculator gets 2 + 2 right but 5 * 5 wrong. Does it work?

Answer: A. No; one wrong result shows a problem

One failure matters.

- ✗ B. It cannot be known: it can
- ✗ C. Yes, mostly: still broken
- ✗ D. Yes, because getting the first problem right proves that every other answer is right: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-B1-Q3 · question page 1794

AI.CODE.02 · Compare checking code with checking a calculator you built

Nice-looking buttons prove what?

Answer: D. Nothing about correctness

Looks are not tests.

- ✗ A. That every answer the calculator gives will always be correct for any problem: no
- ✗ B. That it is cheap: no
- ✗ C. That it is fast: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-B1-Q4 · question page 1795

AI.CODE.02 · Compare checking code with checking a calculator you built

Generated code is like what in this comparison?

Answer: B. A new calculator to test

Needs testing.

- ✗ A. A broken calculator: not necessarily
- ✗ C. A math book: no
- ✗ D. A calculator that has already been fully tested by thousands of people over many years: not yet tested

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-B1-Q5 · question page 1796

AI.CODE.02 · Compare checking code with checking a calculator you built

What should you test code with?

Answer: D. Inputs whose correct answers you know

Known-answer tests.

- ✗ A. Inputs chosen at random whose correct answers nobody knows, so the test is fair: cannot compare
- ✗ B. Only one input: not enough
- ✗ C. No inputs: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F1-Q1 · question page 1797

AI.CODE.02 · Check generated code in practice

A tool writes a function to count vowels. Which test helps most?

Answer: C. Run it on "banana" and expect 3

A known answer: a, a, a.

- ✗ A. Check it has a good name: not a test
- ✗ B. Count its lines: not a test
- ✗ D. Read the comment at the top: not a test

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F1-Q2 · question page 1798

AI.CODE.02 · Check generated code in practice

Generated code calls a function the tool named "sort_smart," which does not exist. What will happen?

Answer: A. An error when it runs

Invented functions fail.

- ✗ B. The computer will create the missing function automatically, based on its name: no
- ✗ C. It will run normally: it does not exist
- ✗ D. It will print the name: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F1-Q3 · question page 1799

AI.CODE.02 · Check generated code in practice

A function to find the largest number works on [3, 9, 2]. What else should you test?

Answer: B. An empty list and negative numbers

Test boundaries.

✗ A. The same list again: no new information

✗ C. Nothing else, since working on one list of three numbers shows it will work on every list: not proof

✗ D. A list of colors: wrong type

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F1-Q4 · question page 1800

AI.CODE.02 · Check generated code in practice

Code will calculate students' grades. What extra step is wise?

Answer: A. Have someone experienced review it

Higher stakes, more review.

✗ B. Make it shorter: not a check

✗ C. Delete the tests: no

✗ D. Use it immediately for every student: affects people

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F1-Q5 · question page 1801

AI.CODE.02 · Check generated code in practice

Which shows that code really works?

Answer: C. Correct results on several known-answer tests

Behavior is evidence.

✗ A. Neat indentation: not proof

✗ B. A long, confident explanation from the tool describing exactly what the code is doing: not proof

✗ D. Many comments: not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F2-Q1 · question page 1802

AI.CODE.02 · Trust behavior over descriptions

A tool explains its code "removes duplicates," but running it leaves duplicates. Which should you trust?

Answer: D. What the code actually does when run

Run it.

- ✗ A. Neither: behavior is evidence
- ✗ B. The explanation, since the tool knows its own code better than any test that you could run: behavior wins
- ✗ C. The comments: can be wrong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F2-Q2 · question page 1803

AI.CODE.02 · Trust behavior over descriptions

A comment says "# adds tax," but the line subtracts. What does this show?

Answer: A. Comments can be wrong

Read the code itself.

- ✗ B. That the subtraction must be correct, because comments are always checked before code runs: no
- ✗ C. That the code will not run: it runs
- ✗ D. That tax is negative: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F2-Q3 · question page 1804

AI.CODE.02 · Trust behavior over descriptions

Which inputs test the boundaries of a function that handles ages?

Answer: D. 0, a very large age, and a missing age

Smallest, largest, missing.

- ✗ A. Only 25: one value
- ✗ B. Only names: wrong type
- ✗ C. Three typical adult ages, such as 30, 35 and 40, which cover the most common situations: not boundaries

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F2-Q4 · question page 1805

AI.CODE.02 · Trust behavior over descriptions

Why do confident explanations make wrong code more convincing?

Answer: B. They make you less likely to test it

Tone discourages checking.

- ✗ A. Because they contain numbers: not the point
- ✗ C. Because confident explanations actually change the code so that it works correctly: no
- ✗ D. Because they are longer: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S86-F2-Q5 · question page 1806

AI.CODE.02 · Trust behavior over descriptions

Code will handle customer payments. What should happen before use?

Answer: C. Experienced review and testing

Money needs review.

- ✗ A. Only a spell check: not enough
- ✗ B. Only a quick glance: not enough

✗ D. Nothing, since payment code written by a tool is always reviewed by a bank automatically: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-M-Q1 · question page 1808

AI.CODE.03 · State requirements clearly, including edge cases

What is a requirement?

Answer: C. A checkable statement of what code must do

Clear, checkable.

✗ A. A guess about what users want: not checkable
✗ B. The program's file name: no
✗ D. A list of the programming languages: not a requirement

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-M-Q2 · question page 1809

AI.CODE.03 · State requirements clearly, including edge cases

Which is an edge case for a test-score average?

Answer: A. An empty list of scores

Empty input often breaks code.

✗ B. A list of ten scores: typical
✗ C. Scores of 80 and 85: typical
✗ D. A typical list: typical case

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-M-Q3 · question page 1810

AI.CODE.03 · State requirements clearly, including edge cases

In this example, which input would break the function?

Prepared code example (Python 3). Shown only; nothing runs.

```
def first_letter(name):
    return name[0]

print(first_letter("Maya"))
```

Answer: B. An empty name ""

name[0] fails when there is no first letter.

- ✗ A. "Zoe": works
- ✗ C. "Al": works
- ✗ D. "Maya," since names with four letters are too long for this function to handle correctly: it works

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S87-M-Q4 · question page 1811

AI.CODE.03 · State requirements clearly, including edge cases

Which is a boundary value for scores from 0 to 100?

Answer: D. 100

Exactly at the limit.

- ✗ A. 72: middle
- ✗ B. 85: middle
- ✗ C. 57: not a boundary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-M-Q5 · question page 1812

AI.CODE.03 · State requirements clearly, including edge cases

Which request includes an edge-case rule?

Answer: A. "If the list is empty, return 0 and do not crash."

Says what to do in an unusual case.

- ✗ B. "Use good names.": no edge case
- ✗ C. "Make it fast.": no edge case
- ✗ D. "Write the code for the average quickly, and make it look clean and professional.": no edge case

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-B1-Q1 · question page 1813

AI.CODE.03 · Compare edge cases with testing a door lock

Why does a locksmith try the wrong key?

Answer: B. To see how the lock handles unusual situations

Test unusual cases.

- ✗ A. To save time: no
- ✗ C. To damage the lock: no
- ✗ D. Because the right key never works on any new lock: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-B1-Q2 · question page 1814

AI.CODE.03 · Compare edge cases with testing a door lock

"No key at all" is like which code input?

Answer: C. An empty input

Nothing provided.

- ✗ A. A very large number: different

✗ B. A correct name: no

✗ D. A typical input: not unusual

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-B1-Q3 · question page 1815

AI.CODE.03 · Compare edge cases with testing a door lock

Freezing weather is like what in code?

Answer: D. Unusual conditions

Stress conditions.

✗ A. A comment: no

✗ B. A variable name: no

✗ C. Normal use: not normal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-B1-Q4 · question page 1816

AI.CODE.03 · Compare edge cases with testing a door lock

Where do locks and code most often fail?

Answer: A. In unusual situations

Edge cases.

✗ B. Never: they fail

✗ C. Only on weekends: no

✗ D. In the most common situation, the one that the designer tested most carefully: least likely

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-B1-Q5 · question page 1817

AI.CODE.03 · Compare edge cases with testing a door lock

What is the lesson for code?

Answer: D. Test unusual inputs, not just the normal one

Edge cases matter.

- ✗ A. Only test long inputs: too narrow
- ✗ B. Never test code: no
- ✗ C. Test only the normal input once, because unusual inputs never happen in real life: they happen

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F1-Q1 · question page 1818

AI.CODE.03 · Write requirements with edge cases in practice

Ben asks: "Write code to split a bill." What key detail is missing?

Answer: C. How many people and how to round

Inputs and rounding rules.

- ✗ A. The restaurant's address: irrelevant
- ✗ B. The programmer's name: irrelevant
- ✗ D. Which color to use for the numbers: not key

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F1-Q2 · question page 1819

AI.CODE.03 · Write requirements with edge cases in practice

A function divides a total by the number of people. Which edge case matters most?

Answer: B. Zero people

Dividing by zero crashes.

✗ A. Three people: typical

✗ C. Two people: typical

✗ D. Four people: typical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F1-Q3 · question page 1820

AI.CODE.03 · Write requirements with edge cases in practice

Which requirement is clearest?

Answer: D. "Return the total price including 8% tax, rounded to cents."

Specific and checkable.

✗ A. "Do prices.": vague

✗ B. "Handle tax.": vague

✗ C. "Make the price thing work properly with tax and everything else.": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F1-Q4 · question page 1821

AI.CODE.03 · Write requirements with edge cases in practice

A form asks for age. Which input is a wrong-type edge case?

Answer: B. "twelve" typed as a word

Text where a number is expected.

✗ A. 15: typical

✗ C. 30: typical

✗ D. 12, typed as a whole number: typical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F1-Q5 · question page 1822

AI.CODE.03 · Write requirements with edge cases in practice

A score must be from 0 to 100. Which two inputs test the boundaries?

Answer: C. 0 and 100

Exactly at the limits.

- ✗ A. 60 and 61: not boundaries
- ✗ B. 20 and 30: not boundaries
- ✗ D. 50 and 75: not boundaries

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F2-Q1 · question page 1823

AI.CODE.03 · Fill gaps in requirements before coding

"Sort the names" leaves which question open?

Answer: A. Alphabetical or by length?

Unstated rules get guessed.

- ✗ B. Which font to show: irrelevant
- ✗ C. Which keyboard to use: irrelevant
- ✗ D. Nothing: not settled

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F2-Q2 · question page 1824

AI.CODE.03 · Fill gaps in requirements before coding

Why write input examples with expected outputs?

Answer: B. They clarify the requirement and become tests

Examples double as tests.

✗ A. Because tools need exactly ten examples: no

✗ C. To make the request longer: not the point

✗ D. Because examples make the code run faster: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F2-Q3 · question page 1825

AI.CODE.03 · Fill gaps in requirements before coding

You are unsure whether "Ana" and "ana" count as the same name. What should you do?

Answer: D. Decide the rule before coding

Choose the rule first.

✗ A. Delete one of the names: not a rule

✗ B. Ignore the question: leads to bugs

✗ C. Let the tool decide quietly: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F2-Q4 · question page 1826

AI.CODE.03 · Fill gaps in requirements before coding

What happens when a requirement is unstated?

Answer: A. The tool fills it with a guess

Guesses may not fit.

✗ B. The tool always asks you a question first and never guesses anything on its own: not always

✗ C. The code refuses to run: not usually

✗ D. Nothing at all: something is chosen

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S87-F2-Q5 · question page 1827

AI.CODE.03 · Fill gaps in requirements before coding

Which example helps clarify "remove extra spaces"?

Answer: C. " hi there " becomes "hi there"

Concrete input and output.

- ✗ A. "hello" stays "hello": no spaces involved
- ✗ B. "Spaces are bad": not an example
- ✗ D. "Make the text look tidier and cleaner": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-M-Q1 · question page 1829

AI.CODE.04 · Choose a test whose expected result would reveal a defect

What does a test compare?

Answer: D. The code's result with an expected result you worked out

Actual versus expected.

- ✗ A. The code's speed with the internet speed: no
- ✗ B. Two programmers' opinions: not a test
- ✗ C. The code's length with the length of another program: not a test

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-M-Q2 · question page 1830

AI.CODE.04 · Choose a test whose expected result would reveal a defect

In this example, which test reveals the bug?

Prepared code example (Python 3). Shown only; nothing runs.

```
def is_adult(age):
```

```
    return age > 18

print(is_adult(20))
print(is_adult(18))
```

Answer: A. is_adult(18)

18 should be True but gives False.

- ✗ B. is_adult(50): passes
- ✗ C. is_adult(30): passes
- ✗ D. is_adult(20): it passes

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S88-M-Q3 · question page 1831

AI.CODE.04 · Choose a test whose expected result would reveal a defect

What is the bug in the same example?

Prepared code example (Python 3). Shown only; nothing runs.

```
def is_adult(age):
    return age > 18

print(is_adult(20))
print(is_adult(18))
```

Answer: B. It uses > instead of >=

18 or older needs >=.

- ✗ A. It prints too much: not the bug
- ✗ C. It has no name: it does
- ✗ D. It uses the word "age,": allowed

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S88-M-Q4 · question page 1832

AI.CODE.04 · Choose a test whose expected result would reveal a defect

What makes a test good?

Answer: C. It would fail if a particular mistake were present

Targeted at a mistake.

- ✗ A. It is very long: not the point
- ✗ B. It uses random inputs: not targeted
- ✗ D. It always passes, which shows that the code is working correctly every single time: not useful

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-M-Q5 · question page 1833

AI.CODE.04 · Choose a test whose expected result would reveal a defect

To catch a mistake at a limit, what should you test?

Answer: B. The limit itself

Boundaries expose errors.

- ✗ A. Nothing: must test
- ✗ C. Only typical values: misses it
- ✗ D. A value far away: misses it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-B1-Q1 · question page 1834

AI.CODE.04 · Pick the input that exposes a suspected mistake

Which input exposes > 50 versus >= 50?

Answer: C. 50

Only 50 differs.

- ✗ A. 30: same either way
- ✗ B. 100: same either way
- ✗ D. 80: same either way

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-B1-Q2 · question page 1835

AI.CODE.04 · Pick the input that exposes a suspected mistake

For input 50, what should the test expect?

Answer: A. Free shipping

"50 or more" includes 50.

- ✗ B. An error: no
- ✗ C. Half-price shipping: no
- ✗ D. Not free: rule says or more

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-B1-Q3 · question page 1836

AI.CODE.04 · Pick the input that exposes a suspected mistake

Why do 30 and 80 not help?

Answer: D. Both rules give the same answer for them

No difference to detect.

- ✗ A. They crash the code: no
- ✗ B. They are too large: no
- ✗ C. Because 30 and 80 are not real prices: they are fine

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-B1-Q4 · question page 1837

AI.CODE.04 · Pick the input that exposes a suspected mistake

What question drives the method?

Answer: C. Which input gives a different answer if the mistake exists?

Target the difference.

- ✗ A. Which input is easiest to type?: not the question
- ✗ B. Which input is the most common?: not the question
- ✗ D. Which input is the largest number that the program is able to handle without crashing?: not the question

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-B1-Q5 · question page 1838

AI.CODE.04 · Pick the input that exposes a suspected mistake

Rule: "under 13 pays child price." Suspected mistake: <= 13. Which input exposes it?

Answer: D. 13

Only 13 differs.

- ✗ A. 12: same either way
- ✗ B. 5, because young children are the main customers who pay the child price at the door: same either way
- ✗ C. 20: same either way

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F1-Q1 · question page 1839

AI.CODE.04 · Design tests that catch mistakes

A function should give a discount to orders of 100 dollars or more. Which test is most useful?

Answer: A. An order of exactly 100, expecting a discount

Test the boundary.

- ✗ B. An order of 500, which is so large that it should definitely get the discount in any case: misses the limit
- ✗ C. No test: must test
- ✗ D. An order of 20, expecting no discount: misses the limit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F1-Q2 · question page 1840

AI.CODE.04 · Design tests that catch mistakes

A function should count words. Which test could reveal a mistake with extra spaces?

Answer: B. "hi there" (two spaces), expecting 2

Target the suspected mistake.

- ✗ A. "hello," a single word with no spaces at all, which is the simplest possible test: no spaces
- ✗ C. An empty test: not targeted
- ✗ D. "one two" with one space: typical

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F1-Q3 · question page 1841

AI.CODE.04 · Design tests that catch mistakes

Ana writes a test expecting the average of [2, 4] to be 3. The code returns 3. What does this show?

Answer: D. The code passes this test

One pass, not full proof.

- ✗ A. That the test is wrong: it is right

✗ B. Nothing at all: it shows a pass

✗ C. That the code is correct: one test

Arithmetic checked by code: yes. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F1-Q4 · question page 1842

AI.CODE.04 · Design tests that catch mistakes

A function should reject passwords shorter than 8 characters. Which input tests the boundary?

Answer: B. A password of exactly 7 characters, expecting rejection

Just below the limit.

✗ A. A password of 30 characters, since longer passwords are the most important ones to check: far from limit

✗ C. A password of 15 characters: far from limit

✗ D. No password: different case

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F1-Q5 · question page 1843

AI.CODE.04 · Design tests that catch mistakes

Which expected result should a test use?

Answer: C. One you worked out yourself

Independent expected answers.

✗ A. The tool's guess: not independent

✗ B. Whatever the code printed the first time it ran, since the code is the best source of answers: circular

✗ D. A random number: not a check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F2-Q1 · question page 1844

AI.CODE.04 · Spot weak tests

A test's expected answer was copied from the code's output. What is the problem?

Answer: A. It only checks the code agrees with itself

Not independent.

- ✗ B. The test uses numbers: not the problem
- ✗ C. Nothing, because the code's own output is always the most reliable expected answer: circular
- ✗ D. The test is too long: not the problem

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F2-Q2 · question page 1845

AI.CODE.04 · Spot weak tests

A test checks only that the result "is a number." Why is it weak?

Answer: C. It passes even when the number is wrong

Checks too little.

- ✗ A. Because numbers are not results: they are
- ✗ B. Because it is too strict: too loose
- ✗ D. Because numbers can never be checked by tests: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F2-Q3 · question page 1846

AI.CODE.04 · Spot weak tests

All tests use typical inputs. What might they miss?

Answer: B. Bugs at boundaries and special cases

Untested paths.

- ✗ A. Spelling mistakes in comments: not the point
- ✗ C. Nothing, since typical inputs exercise every single line of code in every program: not always
- ✗ D. The program's file name: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F2-Q4 · question page 1847

AI.CODE.04 · Spot weak tests

Which test is strongest for a function that returns the largest of three numbers?

Answer: D. Largest in last place: (1, 2, 9) expecting 9

Specific expected answer, tricky position.

- ✗ A. No test: must test
- ✗ B. (5, 5, 5) expecting a result: checks too little
- ✗ C. (9, 2, 1) expecting a number of some kind: checks too little

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S88-F2-Q5 · question page 1848

AI.CODE.04 · Spot weak tests

Where should strong tests focus?

Answer: A. Boundaries and special cases, with independent answers

Targeted and independent.

- ✗ B. Only the program name: no
- ✗ C. Random inputs with no expected answers: cannot compare
- ✗ D. The most common input only, run many times in a row: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S89-M-Q1 · question page 1850

AI.CODE.05 · Use an error message to locate a problem

Where should you start reading a traceback?

Answer: A. At the bottom

The last line names the error.

- ✗ B. In the middle: no
- ✗ C. Nowhere; ignore it: it helps
- ✗ D. At the very top: it does not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-M-Q2 · question page 1851

AI.CODE.05 · Use an error message to locate a problem

In this example, what type of error is shown?

Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
scores = [90, 85, 77]
print(scores[3])

Traceback (most recent call last):
File "grades.py", line 2, in <module>
print(scores[3])
IndexError: list index out of range
```

Answer: B. IndexError

Position 3 does not exist.

- ✗ A. TypeError: no

✗ C. ValueError: no

✗ D. NameError: it was

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-M-Q3 · question page 1852

AI.CODE.05 · Use an error message to locate a problem

In the same example, why is position 3 out of range?

Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
scores = [90, 85, 77]
print(scores[3])

Traceback (most recent call last):
File "grades.py", line 2, in <module>
print(scores[3])
IndexError: list index out of range
```

Answer: C. Python counts positions from 0, so they are 0 to 2

Three items: positions 0, 1, 2.

✗ A. Because 77 is too large: no

✗ B. Because print cannot show lists: no

✗ D. Because lists in Python can only hold two items: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-M-Q4 · question page 1853

AI.CODE.05 · Use an error message to locate a problem

In this example, what caused the NameError?

Prepared code example (Python 3) and its error message, simplified. Shown only;

nothing runs.

```
total = 50
print(totl)

Traceback (most recent call last):
File "shop.py", line 2, in <module>
print(totl)
NameError: name 'totl' is not defined
```

Answer: D. A typo: totl instead of total

A name never defined.

- ✗ A. print is misspelled: it is not
- ✗ B. The file name is wrong: no
- ✗ C. The number 50 is too large: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-M-Q5 · question page 1854

AI.CODE.05 · Use an error message to locate a problem

When asking a tool to fix an error, what should you include?

Answer: B. The full error message

Full message, plus the code.

- ✗ A. Only the words "it is broken," since the tool can work out the rest by itself: too little
- ✗ C. Nothing: too little
- ✗ D. Only the file name: too little

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-B1-Q1 · question page 1855

AI.CODE.05 · Read the last two lines of a traceback

In the example, which file has the error?

Answer: C. pay.py

Named in the File line.

- ✗ A. line 7: that is the line
- ✗ B. ZeroDivisionError, which is the name of the file where the problem happened: that is the type
- ✗ D. rate: that is a variable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-B1-Q2 · question page 1856

AI.CODE.05 · Read the last two lines of a traceback

Which line number?

Answer: A. 7

From the File line.

- ✗ B. 0, because the error message mentions division by zero in the last line: not a line number
- ✗ C. 1: no
- ✗ D. 2: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-B1-Q3 · question page 1857

AI.CODE.05 · Read the last two lines of a traceback

What is the error type?

Answer: D. ZeroDivisionError

The last line.

✗ A. hours: no

✗ B. line 7: no

✗ C. pay.py, which is listed just above the line of code that caused the problem: that is the file

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-B1-Q4 · question page 1858

AI.CODE.05 · Read the last two lines of a traceback

What is the problem?

Answer: B. Dividing by zero

Division by zero.

✗ A. Adding text to numbers: no

✗ C. A missing file: no

✗ D. A misspelled variable name: not this one

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-B1-Q5 · question page 1859

AI.CODE.05 · Read the last two lines of a traceback

Which two lines hold most of the information?

Answer: A. The last two

Read from the bottom.

✗ B. None: they help

✗ C. The middle two: no

✗ D. The first two: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F1-Q1 · question page 1860

AI.CODE.05 · Locate problems from error messages

In this example, what is the problem?

Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
age = 15
print("Age: " + age)

Traceback (most recent call last):
File "card.py", line 2, in <module>
print("Age: " + age)
TypeError: can only concatenate str (not "int") to str
```

Answer: D. Adding text to a number

TypeError: str plus int.

- ✗ A. The variable name "age" is not allowed in Python, so the program cannot run at all: allowed
- ✗ B. A missing file: no
- ✗ C. The number 15 is too large: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F1-Q2 · question page 1861

AI.CODE.05 · Locate problems from error messages

In the same example, which line has the problem?

Prepared code example (Python 3) and its error message, simplified. Shown only; nothing runs.

```
age = 15
print("Age: " + age)
```

```
Traceback (most recent call last):
File "card.py", line 2, in <module>
print("Age: " + age)
TypeError: can only concatenate str (not "int") to str
```

Answer: C. Line 2

Named in the traceback.

- ✗ A. Line 3: there is none
- ✗ B. Line 15: that is the age
- ✗ D. Line 1: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F1-Q3 · question page 1862

AI.CODE.05 · Locate problems from error messages

int("ten") gives "ValueError: invalid literal for int()." What does it mean?

Answer: D. The text is not a number Python can convert

Right type, wrong content.

- ✗ A. The list is too short: IndexError
- ✗ B. The file is missing: no
- ✗ C. The program forgot to define a variable called "ten" before using it on this line: that is NameError

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F1-Q4 · question page 1863

AI.CODE.05 · Locate problems from error messages

A NameError names "prnt." What is the likely cause?

Answer: C. A typo for print

Misspelled name.

- ✗ A. Text plus a number: no
- ✗ B. Dividing by zero: no
- ✗ D. A missing list item: IndexError

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F1-Q5 · question page 1864

AI.CODE.05 · Locate problems from error messages

Before asking a tool for a fix, what is a good first step?

Answer: A. Find the line and error type yourself

Locate the problem.

- ✗ B. Restart the computer: not a fix
- ✗ C. Ignore the message: it helps
- ✗ D. Delete the whole program and start again: unnecessary

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F2-Q1 · question page 1865

AI.CODE.05 · Trace errors to their cause and confirm fixes

An error appears on line 20, but the wrong value was created on line 5. Where is the cause?

Answer: B. Line 5

Errors can surface later.

- ✗ A. Nowhere: there is a cause
- ✗ C. Line 1: not stated
- ✗ D. Line 20: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F2-Q2 · question page 1866

AI.CODE.05 · Trace errors to their cause and confirm fixes

A tool "fixes" an error by making the program ignore all errors silently. What is wrong?

Answer: D. It hides the problem

Address the cause.

- ✗ A. It uses Python: not the issue
- ✗ B. It is too short: not the issue
- ✗ C. Nothing: not true

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F2-Q3 · question page 1867

AI.CODE.05 · Trace errors to their cause and confirm fixes

After applying a fix, what should you do?

Answer: A. Rerun the original failing case

Confirm the fix.

- ✗ B. Delete the test: no
- ✗ C. Ask for another fix: test first
- ✗ D. Move on without testing, since the tool already confirmed the fix would definitely work: confirm

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F2-Q4 · question page 1868

AI.CODE.05 · Trace errors to their cause and confirm fixes

The error is gone, but the output is now wrong. Is it fixed?

Answer: C. No; the result must also be right

No error is not the same as correct.

- ✗ A. Yes, if it is fast: no
- ✗ B. Yes, because removing an error message is the same thing as fixing the problem: not the same
- ✗ D. It cannot be known: it can

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S89-F2-Q5 · question page 1869

AI.CODE.05 · Trace errors to their cause and confirm fixes

Which fix addresses the cause of IndexError from scores[3] on a 3-item list?

Answer: B. Use a valid position, such as scores[2]

Fix the position.

- ✗ A. Hide the error message so that the program keeps running and prints nothing at all: hides it
- ✗ C. Rename the list: not the cause
- ✗ D. Add more print lines: not a fix

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-M-Q1 · question page 1871

AI.CODE.06 · Use explanations as a starting point and review code against requirements

How should you treat a tool's explanation of code?

Answer: C. As a starting point to check

Helpful, but verify.

- ✗ A. As useless: it helps
- ✗ B. As a replacement for running it: no

✗ D. As proof of what the code does: not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-M-Q2 · question page 1872

AI.CODE.06 · Use explanations as a starting point and review code against requirements

In this example, what does the program print?

Prepared code example (Python 3). Shown only; nothing runs.

```
# Return the largest number in the list
def biggest(nums):
    return min(nums)

print(biggest([4, 9, 2]))
```

Answer: A. 2

min() returns the smallest.

✗ B. 4: no

✗ C. 9, because the comment says the function returns the largest number in the list: comment is wrong

✗ D. 15: no sum

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S90-M-Q3 · question page 1873

AI.CODE.06 · Use explanations as a starting point and review code against requirements

In the same example, what is wrong?

Prepared code example (Python 3). Shown only; nothing runs.

```
# Return the largest number in the list
def biggest(nums):
    return min(nums)
```

```
print(biggest([4, 9, 2]))
```

Answer: D. The code uses min() but the requirement is the largest

Code does not match the requirement.

- ✗ A. The comment uses the word "largest," which is not allowed in Python comments: allowed
- ✗ B. Nothing: it is wrong
- ✗ C. The list is too short: no

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S90-M-Q4 · question page 1874

AI.CODE.06 · Use explanations as a starting point and review code against requirements

What does a code review check?

Answer: B. Each line against the requirements

Line by line, against requirements.

- ✗ A. Only the file name: no
- ✗ C. Only the indentation: no
- ✗ D. Only that the code has comments: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-M-Q5 · question page 1875

AI.CODE.06 · Use explanations as a starting point and review code against requirements

What can be wrong in code besides the statements?

Answer: A. Comments, names and explanations

All can mislead.

✗ B. Only the spaces: no

✗ C. Only the colors: no

✗ D. Nothing; only the statements: others can mislead

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-B1-Q1 · question page 1876

AI.CODE.06 · Compare reviewing code with proofreading a recipe

Why not stop at the recipe title?

Answer: B. It says what it should be, not what the lines do

Check the lines.

✗ A. Because cakes have no titles: no

✗ C. Because titles are too short: not the point

✗ D. Because titles of recipes are always wrong: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-B1-Q2 · question page 1877

AI.CODE.06 · Compare reviewing code with proofreading a recipe

The recipe title is like what in code?

Answer: C. A comment

Intent, not behavior.

✗ A. An error message: no

✗ B. A variable: no

✗ D. A test that runs the code on several inputs and compares the results with answers: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-B1-Q3 · question page 1878

AI.CODE.06 · Compare reviewing code with proofreading a recipe

Checking "cup" versus "tablespoon" is like what?

Answer: D. Checking each line's details

Line-by-line review.

- ✗ A. Reading the title of the recipe twice to make sure that the cake is called the right thing: not detail
- ✗ B. Choosing a font: no
- ✗ C. Counting pages: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-B1-Q4 · question page 1879

AI.CODE.06 · Compare reviewing code with proofreading a recipe

Noticing missing eggs is like what in a code review?

Answer: A. Finding something missing

Look for what is not there.

- ✗ B. Renaming a variable: no
- ✗ C. Running the code: no
- ✗ D. Finding a misspelled comment: not missing

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-B1-Q5 · question page 1880

AI.CODE.06 · Compare reviewing code with proofreading a recipe

What decides whether the cake will work?

Answer: D. The actual lines

Behavior comes from statements.

- ✗ A. The title of the recipe, which tells the baker exactly how the cake will turn out: not the title
- ✗ B. The photo: no
- ✗ C. The page color: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F1-Q1 · question page 1881

AI.CODE.06 · Review code against requirements in practice

Requirement: "Return the number of students who passed (score 60 or more)." The code counts scores above 60. What is the issue?

Answer: C. It misses students who scored exactly 60

Above 60 excludes 60.

- ✗ A. It uses numbers: no
- ✗ B. It counts too many: too few
- ✗ D. Nothing, since a score of exactly 60 is so rare: requirement includes it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F1-Q2 · question page 1882

AI.CODE.06 · Review code against requirements in practice

A tool explains code as "sorts names alphabetically," but output shows them by length. What should you trust?

Answer: B. The output

Behavior is evidence.

- ✗ A. Neither: output is evidence
- ✗ C. The comment: can be wrong
- ✗ D. The explanation: behavior wins

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F1-Q3 · question page 1883

AI.CODE.06 · Review code against requirements in practice

During review, you find a line whose purpose you cannot tell. What should you do?

Answer: A. Ask about it and test it before approving

Unclear code is a finding.

- ✗ B. Approve it anyway, since lines you do not understand are usually the most clever parts of code: risky
- ✗ C. Delete it without checking: may break things
- ✗ D. Ignore it: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F1-Q4 · question page 1884

AI.CODE.06 · Review code against requirements in practice

Which review question asks about missing parts?

Answer: D. "Is anything the requirement needs not here?"

Look for omissions.

- ✗ A. "Is the font readable?": no
- ✗ B. "Is it short?": no
- ✗ C. "Is the variable name long enough to describe exactly what the variable is used for?": naming

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F1-Q5 · question page 1885

AI.CODE.06 · Review code against requirements in practice

Which is the best use of a code explanation?

Answer: C. To learn what to check, then verify by running

Starting point.

- ✗ A. To decide the requirement: requirements come first
- ✗ B. To skip reading the code: no
- ✗ D. To replace testing entirely, since a clear explanation proves that the code is correct: not proof

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F2-Q1 · question page 1886

AI.CODE.06 · Review for what is missing and keep changes reviewable

Code divides by the number of items but never checks for zero items. What kind of finding is this?

Answer: B. A missing check

Something absent.

- ✗ A. A comment problem: no
- ✗ C. No problem: it can crash
- ✗ D. A naming problem: not naming

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F2-Q2 · question page 1887

AI.CODE.06 · Review for what is missing and keep changes reviewable

A tool proposes changing 400 lines at once. What helps review?

Answer: A. Asking for smaller changes

Small changes are reviewable.

✗ B. Reading only the first line: not enough

✗ C. Skipping review: risky

✗ D. Approving all 400 lines quickly: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F2-Q3 · question page 1888

AI.CODE.06 · Review for what is missing and keep changes reviewable

You cannot understand part of a generated function. What is the right response?

Answer: C. Ask for a simpler version or verify with small examples

Understanding is part of review.

✗ A. Pretend you understand: no

✗ B. Remove the whole program: excessive

✗ D. Approve it, because code that is hard to understand is usually more efficient and better: not a reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F2-Q4 · question page 1889

AI.CODE.06 · Review for what is missing and keep changes reviewable

A requirement says "log every failed login," but the code logs only some. What is this?

Answer: B. A skipped requirement

Missing behavior.

✗ A. A faster design: not the point

✗ C. A typo: no

✗ D. A style choice: not optional

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S90-F2-Q5 · question page 1890

AI.CODE.06 · Review for what is missing and keep changes reviewable

Which review finding is most serious?

Answer: D. An unhandled error that can crash the program

Crashes matter.

- ✗ A. A blank line: minor
- ✗ B. A long variable name: minor
- ✗ C. A comment that uses slightly informal language: minor

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-M-Q1 · question page 1892

AI.CODE.07 · Explain commits, diffs and reverting

What is a commit?

Answer: B. A saved snapshot of code with a message

Snapshot plus message.

- ✗ A. A promise from the programmer to finish the project by a certain date in the future: not the meaning
- ✗ C. An error message: no
- ✗ D. A deleted file: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-M-Q2 · question page 1893

AI.CODE.07 · Explain commits, diffs and reverting

In a diff, what does a line starting with "+" mean?

Answer: A. It was added

Plus means added.

- ✗ B. It is a comment: no
- ✗ C. It has an error: no
- ✗ D. It was removed: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-M-Q3 · question page 1894

AI.CODE.07 · Explain commits, diffs and reverting

In this example, what changed?

Prepared example (a diff in a fictional project). Shown only; nothing runs.

```
- total = price * count
+ total = price + count
```

Answer: C. Multiplication became addition

"*" removed, "+" added.

- ✗ A. The variable was renamed: no
- ✗ B. Nothing changed: a line changed
- ✗ D. Addition became multiplication: reversed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-M-Q4 · question page 1895

AI.CODE.07 · Explain commits, diffs and reverting

What does reverting do?

Answer: D. Undoes a commit with the opposite change

Undo a change.

✗ A. Copies the project: no

✗ B. Runs the tests: no

✗ C. Deletes the whole project and its history: not reverting

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-M-Q5 · question page 1896

AI.CODE.07 · Explain commits, diffs and reverting

Why commit before accepting generated changes?

Answer: B. To see what changed and undo it if needed

A safe restore point.

✗ A. To hide changes: no

✗ C. To make the code run faster: no

✗ D. Because tools refuse to change code: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-B1-Q1 · question page 1897

AI.CODE.07 · Compare version control with save points in a video game

A game save is like what?

Answer: C. A commit

A snapshot.

✗ A. A test: no

✗ B. A bug: no

✗ D. A diff: not a diff

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-B1-Q2 · question page 1898

AI.CODE.07 · Compare version control with save points in a video game

Comparing two saves to see what changed is like what?

Answer: A. A diff

What changed.

- ✗ B. A new game: no
- ✗ C. A revert: no
- ✗ D. A commit message: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-B1-Q3 · question page 1899

AI.CODE.07 · Compare version control with save points in a video game

Loading an earlier save is like what?

Answer: D. Reverting

Going back.

- ✗ A. Buying a game: no
- ✗ B. Committing a brand-new change that adds a new level to the game for every player: no
- ✗ C. Deleting the game: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-B1-Q4 · question page 1900

AI.CODE.07 · Compare version control with save points in a video game

Saving before a friend plays is like what?

Answer: A. Committing before accepting AI changes

A restore point first.

✗ B. Deleting every save file so that your friend can start the whole game from the beginning: opposite

✗ C. Changing the controls: no

✗ D. Turning off the game: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-B1-Q5 · question page 1901

AI.CODE.07 · Compare version control with save points in a video game

Why save before a hard level?

Answer: C. To return if things go badly

Safety net.

✗ A. Because saving makes the next level easier to beat on the very first try: no

✗ B. To end the game: no

✗ D. To change the music: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F1-Q1 · question page 1902

AI.CODE.07 · Use commits and diffs with AI changes

Ravi lets a tool edit his project and now tests fail. He committed before the edit. What can he do?

Answer: D. Revert to the earlier commit

The commit is a restore point.

✗ A. Delete the project: unnecessary

✗ B. Ignore the tests: risky

✗ C. Nothing, because changes made by a tool can never be undone once they have been accepted: he can revert

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S91-F1-Q2 · question page 1903

AI.CODE.07 · Use commits and diffs with AI changes

A diff shows "- if age >= 18:" and "+ if age > 18:". What changed?

Answer: B. The rule now excludes exactly 18

Boundary changed.

- ✗ A. Nothing important: it matters
- ✗ C. The variable name: no
- ✗ D. The rule now includes more ages: it excludes 18

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F1-Q3 · question page 1904

AI.CODE.07 · Use commits and diffs with AI changes

Which commit message is most useful?

Answer: C. "Fix average to handle an empty list"

What and why.

- ✗ A. "asdf": meaningless
- ✗ B. "stuff": vague
- ✗ D. "Changes," which tells everyone that something in the project was changed at that time: vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F1-Q4 · question page 1905

AI.CODE.07 · Use commits and diffs with AI changes

Why read the diff after a tool edits code?

Answer: B. To see exactly what it changed

Know what changed.

- ✗ A. To rename files: no
- ✗ C. Because diffs fix every bug automatically when they are opened and read by a person: no
- ✗ D. To make the code longer: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F1-Q5 · question page 1906

AI.CODE.07 · Use commits and diffs with AI changes

A diff shows only "+" lines and no "-" lines. What happened?

Answer: A. Lines were only added

Additions only.

- ✗ B. Nothing changed: lines added
- ✗ C. Every line in the file was deleted and replaced with completely different lines: no removals
- ✗ D. Only comments were removed: no removals

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F2-Q1 · question page 1907

AI.CODE.07 · Keep history clear and safe

A diff from a tool edit shows changes to a settings file you did not ask about. What should you do?

Answer: D. Review those changes before accepting

Unexpected changes need review.

- ✗ A. Ignore the diff: risky

✗ B. Delete the settings file: risky

✗ C. Accept everything, since tools only change files that were needed for the requested task: not always

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F2-Q2 · question page 1908

AI.CODE.07 · Keep history clear and safe

Why prefer small commits?

Answer: C. They are easier to review and undo

One thing per commit.

✗ A. Because large commits are not allowed by any version control tool that exists today: not the reason

✗ B. Because they hide changes: no

✗ D. Because small commits use more storage: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F2-Q3 · question page 1909

AI.CODE.07 · Keep history clear and safe

A password was committed and then deleted in the next commit. Is it gone?

Answer: A. No; history still has it

Never commit secrets.

✗ B. Yes, after one day: no

✗ C. Only if the file is small: no

✗ D. Yes: history keeps it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F2-Q4 · question page 1910

AI.CODE.07 · Keep history clear and safe

What should a commit message say?

Answer: B. What changed and why

Clear history.

- ✗ A. Nothing: loses meaning
- ✗ C. Only the programmer's name: recorded already
- ✗ D. Only the date and time: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S91-F2-Q5 · question page 1911

AI.CODE.07 · Keep history clear and safe

Which history is easiest to understand later?

Answer: D. Small commits with clear messages

Clarity for future readers.

- ✗ A. No commits: no history
- ✗ B. Commits with random messages: hard to follow
- ✗ C. One huge commit called "update": hard to follow

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S92-M-Q1 · question page 1913

AI.CODE.08 · Check that a suggested library exists and is trustworthy

What is a package or dependency?

Answer: B. Code written by others that your program installs and uses

Reusable code from others.

✗ A. A box that a computer is shipped in, which contains the cables and the instructions: not the meaning

✗ C. An error message: no

✗ D. A comment: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

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S92-M-Q2 · question page 1914

AI.CODE.08 · Check that a suggested library exists and is trustworthy

What can go wrong when a tool suggests a package?

Answer: A. It may not exist, and attackers can register the name

Invented names can be exploited.

✗ B. Nothing, because tools only ever suggest packages that have been checked for safety: not true

✗ C. It may have documentation: not a risk

✗ D. It may be too popular: not the risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

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S92-M-Q3 · question page 1915

AI.CODE.08 · Check that a suggested library exists and is trustworthy

What is "slopsquatting"?

Answer: D. Registering package names that AI tools invent

Exploiting hallucinated names.

✗ A. Writing very messy code on purpose so that other people find it hard to read and copy: not the meaning

✗ B. Sorting a list: no

✗ C. Deleting old packages: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: code_slopsquat

S92-M-Q4 · question page 1916

AI.CODE.08 · Check that a suggested library exists and is trustworthy

Which is a sign of a trustworthy package?

Answer: C. An active project with real documentation and users

Signs of trust.

- ✗ A. A name that looks like a famous one: warning sign
- ✗ B. No description: warning sign
- ✗ D. A brand-new name that nobody has used, with no documentation: warning signs

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

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S92-M-Q5 · question page 1917

AI.CODE.08 · Check that a suggested library exists and is trustworthy

Before installing a suggested package, what should you check first?

Answer: D. That it exists and is the intended one

Existence and identity.

- ✗ A. Its logo color: no
- ✗ B. Nothing: must check
- ✗ C. That its name is short enough to type quickly: not the check

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

---

S92-B1-Q1 · question page 1918

AI.CODE.08 · Compare package checks with checking a contractor

Why check that a contractor exists?

Answer: C. Because you are letting them into your home

Access requires trust.

- ✗ A. Because contractors who exist always do perfect work, so existence is all that matters: not enough
- ✗ B. To learn their age: no
- ✗ D. To get a discount: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

---

S92-B1-Q2 · question page 1919

AI.CODE.08 · Compare package checks with checking a contractor

A name one letter off from a famous company is like what?

Answer: A. A look-alike package name

Impersonation risk.

- ✗ B. A test: no
- ✗ C. A commit: no
- ✗ D. A well-known package used by thousands of programmers and checked carefully: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

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S92-B1-Q3 · question page 1920

AI.CODE.08 · Compare package checks with checking a contractor

Reviews from other customers are like what for a package?

Answer: B. Signs it is used by others

Evidence of trust.

- ✗ A. Its name length: no
- ✗ C. Its error messages: no

✗ D. The package's file size: not the comparison

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

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S92-B1-Q4 · question page 1921

AI.CODE.08 · Compare package checks with checking a contractor

Installing a package is like what?

Answer: A. Letting someone else's work into your project

Trusting others' code.

✗ B. Deleting a file: no
✗ C. Running a test: no
✗ D. Writing every line of the program yourself: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

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S92-B1-Q5 · question page 1922

AI.CODE.08 · Compare package checks with checking a contractor

What is the lesson?

Answer: D. Check a package before installing it

Verify first.

✗ A. Install anything suggested: risky
✗ B. Only use paid packages: not the point
✗ C. Never use code written by anyone else: too strong

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

---

S92-F1-Q1 · question page 1923

AI.CODE.08 · Check suggested packages in practice

A tool suggests "pip install easy-pdf-magic-tools." You find no project page. What should you do?

Answer: B. Do not install it; look for a well-known alternative

No evidence it is real or safe.

- ✗ A. Rename it and install: no
- ✗ C. Install it twice: no
- ✗ D. Install it quickly, since a missing project page just means the package is very new: warning sign

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

---

S92-F1-Q2 · question page 1924

AI.CODE.08 · Check suggested packages in practice

A suggestion says "reqeusts." What should you notice?

Answer: C. It is a look-alike of "requests"

Misspelled look-alike.

- ✗ A. It is a newer version: not shown
- ✗ B. Nothing, because small spelling differences in package names do not matter at all: they matter
- ✗ D. It is a test file: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

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S92-F1-Q3 · question page 1925

AI.CODE.08 · Check suggested packages in practice

Which package choice is safest for a class project?

Answer: D. One the course already uses

Known and trusted.

✗ A. A random new package: unknown

✗ B. A package that a tool invented a name for, since invented packages are designed for the task: may not exist

✗ C. One with no documentation: warning sign

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

---

S92-F1-Q4 · question page 1926

AI.CODE.08 · Check suggested packages in practice

A tool suggests a package to add two numbers. What is the better choice?

Answer: A. Write the one line yourself

Avoid unnecessary dependencies.

✗ B. Skip adding numbers: no

✗ C. Install three packages: worse

✗ D. Install the package: unneeded risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

---

S92-F1-Q5 · question page 1927

AI.CODE.08 · Check suggested packages in practice

Where should you check that a Python package exists?

Answer: B. Its page on the official package index (PyPI)

Check the official index.

✗ A. In the tool's answer itself, since the tool would not suggest a package that did not exist: circular

✗ C. Your memory: not reliable

✗ D. A random forum: not reliable

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

S92-F2-Q1 · question page 1928

AI.CODE.08 · Manage dependency risks over time

A real, popular package has had no updates for years and has open security warnings. What is the concern?

Answer: C. It may be abandoned

Old, unmaintained risk.

- ✗ A. It has too many users: not the concern
- ✗ B. It is too popular: not the concern
- ✗ D. No concern: can be abandoned

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

---

S92-F2-Q2 · question page 1929

AI.CODE.08 · Manage dependency risks over time

Why record exact package versions?

Answer: D. So builds are repeatable and changes are deliberate

Predictable dependencies.

- ✗ A. To make files longer: no
- ✗ B. To hide which packages are used: no
- ✗ C. Because package managers refuse to install anything: not the reason

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

---

S92-F2-Q3 · question page 1930

AI.CODE.08 · Manage dependency risks over time

A student is unsure whether a package is trustworthy. What is the best step?

Answer: B. Ask someone experienced before installing

Ask first.

- ✗ A. Install it and hope: risky
- ✗ C. Share it with classmates: spreads risk
- ✗ D. Install it on the school's shared computer first: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

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S92-F2-Q4 · question page 1931

AI.CODE.08 · Manage dependency risks over time

Which name is the most suspicious?

Answer: A. "numpy-official-secure-v2"

Extra words can mark look-alikes.

- ✗ B. "requests": well known
- ✗ C. "numpy," which is a widely used package for working with numbers in Python programs: well known
- ✗ D. "pandas": well known

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

---

S92-F2-Q5 · question page 1932

AI.CODE.08 · Manage dependency risks over time

Do real packages ever become unsafe?

Answer: C. Yes; they can be compromised

Ongoing vigilance.

- ✗ A. Only if renamed: no
- ✗ B. No; once a package is real and popular, it can never become unsafe in any way at all: they can
- ✗ D. Only on weekends: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: code_slopsquat

---

S93-M-Q1 · question page 1934

AI.CODE.09 · Spot exposed secrets and unsafe handling of input

What is an API key?

Answer: C. A code that lets a program use a private or paid service

A credential.

- ✗ A. An error message: no
- ✗ B. A variable name: no
- ✗ D. A keyboard key that programmers press to run their code faster: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-M-Q2 · question page 1935

AI.CODE.09 · Spot exposed secrets and unsafe handling of input

In this example, what is the security problem?

Prepared code example (Python 3) with a fake key. Shown only; nothing runs.

```
API_KEY = "sk-live-4f9a2c7e1b"
print("Connecting with key", API_KEY)
```

Answer: D. The key is written directly in the code

Secrets should be stored separately.

- ✗ A. The variable is called API_KEY in capital letters, which Python does not allow: allowed
- ✗ B. Nothing; keys belong in code: they do not
- ✗ C. The code prints a message: not the main problem

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S93-M-Q3 · question page 1936

AI.CODE.09 · Spot exposed secrets and unsafe handling of input

Where should secrets be stored?

Answer: B. Separately, such as in an environment variable

Outside shared code.

- ✗ A. In a comment at the top of the code, where every programmer can find them quickly: still exposed
- ✗ C. In a chat with an AI tool: never
- ✗ D. In the file name: exposed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-M-Q4 · question page 1937

AI.CODE.09 · Spot exposed secrets and unsafe handling of input

In this example, why is the query unsafe?

Prepared code fragment (Python 3). Shown only; nothing runs.

```
name = input("Name: ")
query = "SELECT * FROM students WHERE name = '" + name + "'"
```

Answer: A. Typed text goes straight into the query

Attackers can change the query.

- ✗ B. The table is called students: irrelevant
- ✗ C. It uses the input function: not by itself
- ✗ D. The query is written in capital letters: not the risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-M-Q5 · question page 1938

AI.CODE.09 · Spot exposed secrets and unsafe handling of input

Should you paste a real API key into an AI chat to get help?

Answer: D. No; never paste secrets into a chat

Keep secrets out.

- ✗ A. Yes, if it is short: no
- ✗ B. Only on weekends: no
- ✗ C. Yes, because chats with AI tools are completely private and are never stored anywhere: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-B1-Q1 · question page 1939

AI.CODE.09 · Compare secrets in code with a house key under the doormat

Why is a key under the doormat risky?

Answer: C. Anyone who looks can find it

Easy to find.

- ✗ A. Because keys are heavy: no
- ✗ B. Because doormats damage keys over time, making them stop working in the lock: not the reason
- ✗ D. Because doors are locked: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-B1-Q2 · question page 1940

AI.CODE.09 · Compare secrets in code with a house key under the doormat

A secret written in code is like what?

Answer: B. A key under the doormat

Exposed to anyone who looks.

- ✗ A. A locked safe: no
- ✗ C. A doorbell: no
- ✗ D. A key kept safely: opposite

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-B1-Q3 · question page 1941

AI.CODE.09 · Compare secrets in code with a house key under the doormat

Keeping the key in your pocket is like what?

Answer: A. Storing secrets outside the code

Separate storage.

- ✗ B. Giving it away: no
- ✗ C. Posting it online: no
- ✗ D. Writing the secret into a comment: still exposed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-B1-Q4 · question page 1942

AI.CODE.09 · Compare secrets in code with a house key under the doormat

If the key may have been copied, what should you do?

Answer: C. Change the lock

Like rotating a secret.

- ✗ A. Ignore it: risky
- ✗ B. Paint the door: no
- ✗ D. Move the key: still copied

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-B1-Q5 · question page 1943

AI.CODE.09 · Compare secrets in code with a house key under the doormat

In code, "changing the lock" means what?

Answer: A. Replacing the exposed secret with a new one

Rotate the key.

- ✗ B. Adding a comment: no
- ✗ C. Renaming the variable: not enough
- ✗ D. Deleting the line with the secret from the newest version of the code only: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F1-Q1 · question page 1944

AI.CODE.09 · Handle secrets and input safely in practice

Mia shares her project online and realizes her real API key is in the code. What should she do first?

Answer: D. Change the key at the service right away

Rotate exposed secrets.

- ✗ A. Ignore it: risky
- ✗ B. Delete the key from the code and assume the problem is solved, since nobody saw it yet: may be copied
- ✗ C. Make the repository bigger: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F1-Q2 · question page 1945

AI.CODE.09 · Handle secrets and input safely in practice

A tool asks you to paste your password so it can "test the login code." What should you do?

Answer: B. Refuse; use a fake test password

Use fake test values.

- ✗ A. Paste it, because tools need the real password to check whether the login code works: never needed
- ✗ C. Paste half of it: still risky
- ✗ D. Email it instead: still risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F1-Q3 · question page 1946

AI.CODE.09 · Handle secrets and input safely in practice

Which line handles a secret more safely?

Answer: C. Reading the key from an environment variable

Kept outside the code.

- ✗ A. Printing the key at startup: exposed
- ✗ B. Saving the key in the file name: exposed
- ✗ D. Writing the key in a comment next to the code: exposed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F1-Q4 · question page 1947

AI.CODE.09 · Handle secrets and input safely in practice

A form's code builds a database query by adding typed text. What is the risk?

Answer: D. An attacker can change the query

Unsafe input.

✗ A. Nothing: it is a risk

✗ B. The text will be misspelled: not the risk

✗ C. The query will run more slowly: not the risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F1-Q5 · question page 1948

AI.CODE.09 · Handle secrets and input safely in practice

Generated code contains "API_KEY = 'your-key-here'." What is the safe next step?

Answer: A. Load the real key from a private setting

Keep real keys out of code.

✗ B. Delete the line: code may fail

✗ C. Share the placeholder: not the issue

✗ D. Replace the placeholder text with your real key: exposes it

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F2-Q1 · question page 1949

AI.CODE.09 · Respond to exposed secrets and untrusted input

Why is deleting an exposed key from the code not enough?

Answer: B. It may already have been copied

Rotate the key.

✗ A. Because keys grow back: no

✗ C. Because code cannot be edited: no

✗ D. Because deleting lines of code: possible

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F2-Q2 · question page 1950

AI.CODE.09 · Respond to exposed secrets and untrusted input

What does "rotate a secret" mean?

Answer: D. Replace it with a new one at the service

Issue a new secret.

- ✗ A. Write it backward: no
- ✗ B. Encrypt the file name: no
- ✗ C. Move the secret to a different line: still exposed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F2-Q3 · question page 1951

AI.CODE.09 · Respond to exposed secrets and untrusted input

What do parameterized queries do?

Answer: A. Keep user text separate from the command

Safe input handling.

- ✗ B. Print the query: no
- ✗ C. Remove the database: no
- ✗ D. Make the database query run on a schedule every hour without any user input at all: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F2-Q4 · question page 1952

AI.CODE.09 · Respond to exposed secrets and untrusted input

How should code treat what a user types?

Answer: B. As untrusted

Untrusted input.

✗ A. As a secret: no

✗ C. As a comment: no

✗ D. As fully trusted: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S93-F2-Q5 · question page 1953

AI.CODE.09 · Respond to exposed secrets and untrusted input

A tutorial's generated code prints the API key at startup "for debugging." What is the risk?

Answer: C. The key can appear in logs or screenshots

Do not print secrets.

✗ A. The program runs slower: minor

✗ B. No risk, since printed text disappears from the screen as soon as the program finishes: logs keep it

✗ D. The key gets longer: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-M-Q1 · question page 1955

AI.CODE.10 · Keep generated code understandable, documented and owned

Is code read or written more often?

Answer: A. Read and changed more often

Maintenance dominates.

✗ B. Neither: it is read

✗ C. Written more often, because each line of code is written many times and read only once: reversed

✗ D. Only written: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-M-Q2 · question page 1956

AI.CODE.10 · Keep generated code understandable, documented and owned

In this example, which function is easier to maintain?

Prepared code example (Python 3). Shown only; nothing runs.

```
def f(a, b):
    return a * b * 1.0825

def price_with_tax(subtotal, quantity):
    TAX_RATE = 0.0825  # local sales tax
    return subtotal * quantity * (1 + TAX_RATE)

print(round(f(10, 2), 2), round(price_with_tax(10, 2), 2))
```

Answer: B. price_with_tax

Clear names and an explained rate.

✗ A. Neither: one is clearer

✗ C. f, because short names like f, a and b are faster to type and always easier to understand: unclear

✗ D. Both equally: one is clearer

Arithmetic checked by code: not applicable. Code example run and its output checked: yes. Sources: none (stable)

S94-M-Q3 · question page 1957

AI.CODE.10 · Keep generated code understandable, documented and owned

In the same example, do both functions give the same result?

Prepared code example (Python 3). Shown only; nothing runs.

```
def f(a, b):
    return a * b * 1.0825

def price_with_tax(subtotal, quantity):
```

```
    TAX_RATE = 0.0825  # local sales tax
    return subtotal * quantity * (1 + TAX_RATE)

print(round(f(10, 2), 2), round(price_with_tax(10, 2), 2))
```

Answer: D. Yes, 21.65

Same math, different clarity.

- ✗ A. No; f gives 20: it gives 21.65
- ✗ B. No; one crashes: both run
- ✗ C. No, because functions with different names always calculate different results: same math

Arithmetic checked by code: yes. Code example run and its output checked: yes. Sources: none (stable)

S94-M-Q4 · question page 1958

AI.CODE.10 · Keep generated code understandable, documented and owned

What is a README?

Answer: C. A document explaining what a project does and how to run it

Project documentation.

- ✗ A. A password file: no
- ✗ B. An error log: no
- ✗ D. A file that the computer reads before every program: not the meaning

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-M-Q5 · question page 1959

AI.CODE.10 · Keep generated code understandable, documented and owned

Who should own a piece of code in use?

Answer: D. A person who understands it and fixes it

Clear responsibility.

- ✗ A. Nobody: needs an owner
- ✗ B. The AI tool that wrote it, which will maintain and fix it forever without any help: tools are not owners
- ✗ C. Whoever used it last: not responsible

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-B1-Q1 · question page 1960

AI.CODE.10 · Compare maintainable code with a well-labeled kitchen

Labeled jars are like what in code?

Answer: A. Clear names and comments

Labels help others.

- ✗ B. Error messages: no
- ✗ C. Passwords: no
- ✗ D. Very short names like x and y: unclear

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-B1-Q2 · question page 1961

AI.CODE.10 · Compare maintainable code with a well-labeled kitchen

Unlabeled jars cause what problem?

Answer: B. Only one person knows what is what

Hidden knowledge.

- ✗ A. Food is cheaper: no
- ✗ C. Everyone in the kitchen always knows exactly what is in every jar without looking: opposite
- ✗ D. Jars break: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-B1-Q3 · question page 1962

AI.CODE.10 · Compare maintainable code with a well-labeled kitchen

The note on the fridge is like what?

Answer: C. A README

How things work.

- ✗ A. A commit message that explains a single change made to one line of the code: not the comparison
- ✗ B. A bug: no
- ✗ D. A test: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-B1-Q4 · question page 1963

AI.CODE.10 · Compare maintainable code with a well-labeled kitchen

The person who keeps the kitchen organized is like what?

Answer: B. The code's owner

Responsibility.

- ✗ A. The fridge: no
- ✗ C. A guest: no
- ✗ D. The AI tool: not the owner

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-B1-Q5 · question page 1964

AI.CODE.10 · Compare maintainable code with a well-labeled kitchen

Salt in the cake is like what in code?

Answer: C. A mistake caused by unclear code

Confusion leads to errors.

- ✗ A. A successful new feature that customers asked for and that works exactly as planned: opposite
- ✗ B. A faster program: no
- ✗ D. A new commit: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F1-Q1 · question page 1965

AI.CODE.10 · Keep generated code maintainable in practice

A club uses a generated script to track dues, but nobody can explain how it works. What is the risk?

Answer: A. No one can fix it when it breaks

Unowned code is fragile.

- ✗ B. It uses too little memory: not the risk
- ✗ C. No risk, because generated scripts never break and never need to be changed by anyone: they can
- ✗ D. It will run too fast: not the risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F1-Q2 · question page 1966

AI.CODE.10 · Keep generated code maintainable in practice

Which variable name is clearest?

Answer: D. total_price

Says what it holds.

- ✗ A. thing: unclear

✗ B. tp: unclear

✗ C. x2: unclear

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F1-Q3 · question page 1967

AI.CODE.10 · Keep generated code maintainable in practice

A generated function does five different jobs. What improves it?

Answer: C. Splitting it into smaller functions

One job per function.

✗ A. Renaming it to "do_all": not a fix

✗ B. Removing all comments: worse

✗ D. Adding a sixth job to the same function: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F1-Q4 · question page 1968

AI.CODE.10 · Keep generated code maintainable in practice

Which comment is most useful?

Answer: D. "# Round down because the store never charges partial cents"

Explains why.

✗ A. "# This line of code does something," which reminds readers that the line is important: uninformative

✗ B. "# code": uninformative

✗ C. "# hi": uninformative

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F1-Q5 · question page 1969

AI.CODE.10 · Keep generated code maintainable in practice

Before relying on generated code at work, what should be true?

Answer: B. Someone can explain how it works

Understanding first.

- ✗ A. It should have no comments: no
- ✗ C. It should be secret: no
- ✗ D. It should be long: not a sign

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F2-Q1 · question page 1970

AI.CODE.10 · Prevent quiet maintenance problems

Code contains 0.0825 with no explanation. What is this called?

Answer: A. A magic number

Unexplained value.

- ✗ B. A test: no
- ✗ C. A comment: no
- ✗ D. A secret key: not a key

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F2-Q2 · question page 1971

AI.CODE.10 · Prevent quiet maintenance problems

The same calculation is copied in three places. What problem follows?

Answer: B. A fix in one place may be missed in others

Duplication risk.

- ✗ A. It is easier to read: not usually

✗ C. The program runs faster: not the point

✗ D. Nothing, because copied code always gets updated in every place at the same time: it does not

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F2-Q3 · question page 1972

AI.CODE.10 · Prevent quiet maintenance problems

A tool generates a 300-line solution for a simple task. What should you ask for?

Answer: A. A simpler version

Simpler is easier to own.

✗ B. An even longer version with more features, so it is ready for any future need: harder to maintain

✗ C. Nothing; use it: hard to maintain

✗ D. A version without tests: worse

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F2-Q4 · question page 1973

AI.CODE.10 · Prevent quiet maintenance problems

The student who built the class website is graduating. What should happen?

Answer: C. Hand over knowledge, not just files

Transfer understanding.

✗ A. Change the password and tell no one: loses access

✗ B. Delete the website: unnecessary

✗ D. Nothing: not enough

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S94-F2-Q5 · question page 1974

AI.CODE.10 · Prevent quiet maintenance problems

Why add tests to generated code you will keep?

Answer: D. They describe and protect how it should behave

Tests document behavior.

- ✗ A. To slow it down: no
- ✗ B. To hide bugs: no
- ✗ C. Because tests make the code shorter and remove the need for any comments or README: not the point

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-M-Q1 · question page 1976

AI.AGENT.01 · Tell a chatbot, a fixed workflow and an agent apart

What does a chatbot do?

Answer: C. Answers in a conversation while you decide what to do

You act on its replies.

- ✗ A. Prints documents: no
- ✗ B. Follows a fixed schedule: that is a workflow
- ✗ D. Takes actions in the world on its own, such as sending emails: that is an agent

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-M-Q2 · question page 1977

AI.AGENT.01 · Tell a chatbot, a fixed workflow and an agent apart

What makes a workflow "fixed"?

Answer: D. A person designed its steps in advance

Predetermined steps.

- ✗ A. It never uses a model: it may
- ✗ B. It has no steps: it has steps
- ✗ C. The model chooses each new step by itself: that is an agent

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-M-Q3 · question page 1978

AI.AGENT.01 · Tell a chatbot, a fixed workflow and an agent apart

What makes something an agent?

Answer: B. It chooses its own next steps toward a goal

Goal plus self-chosen steps.

- ✗ A. It has a friendly name: no
- ✗ C. It runs on a timer: that is scheduling
- ✗ D. It answers one question at a time and waits: that is a chatbot

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-M-Q4 · question page 1979

AI.AGENT.01 · Tell a chatbot, a fixed workflow and an agent apart

"When a form arrives, summarize it, then email the manager" describes what?

Answer: A. A tool-using workflow

Fixed steps with a model inside.

- ✗ B. A chatbot: no conversation
- ✗ C. A search engine: no
- ✗ D. An agent: steps are fixed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-M-Q5 · question page 1980

AI.AGENT.01 · Tell a chatbot, a fixed workflow and an agent apart

Which needs the most limits and oversight?

Answer: D. An agent

More freedom, more risk.

- ✗ A. A calculator: no
- ✗ B. A fixed workflow: more predictable
- ✗ C. A chatbot: fewer actions

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-B1-Q1 · question page 1981

AI.AGENT.01 · Compare the three with a GPS, a delivery route and a personal assistant

A GPS you ask for directions is like what?

Answer: B. A chatbot

It suggests; you act.

- ✗ A. A calendar: no
- ✗ C. A workflow: not fixed stops
- ✗ D. An agent: you drive

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-B1-Q2 · question page 1982

AI.AGENT.01 · Compare the three with a GPS, a delivery route and a personal assistant

A fixed delivery route is like what?

Answer: C. A workflow

Designed in advance.

- ✗ A. A map: not the comparison
- ✗ B. A chatbot: no conversation
- ✗ D. An agent: fixed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-B1-Q3 · question page 1983

AI.AGENT.01 · Compare the three with a GPS, a delivery route and a personal assistant

An assistant choosing taxis and trains is like what?

Answer: A. An agent

Goal plus chosen steps.

- ✗ B. A timetable: no
- ✗ C. A fixed workflow: not fixed
- ✗ D. A chatbot: it chooses

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-B1-Q4 · question page 1984

AI.AGENT.01 · Compare the three with a GPS, a delivery route and a personal assistant

Which option is most flexible?

Answer: B. The personal assistant

Chooses its own steps.

- ✗ A. All are equal: they differ
- ✗ C. The GPS: suggests only
- ✗ D. The fixed delivery route: fixed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-B1-Q5 · question page 1985

AI.AGENT.01 · Compare the three with a GPS, a delivery route and a personal assistant

Whose choices most need checking?

Answer: A. The assistant's

Most freedom, most checking.

- ✗ B. The route's: predictable
- ✗ C. No one's: checking matters
- ✗ D. The GPS's: you decide

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F1-Q1 · question page 1986

AI.AGENT.01 · Classify AI setups in practice

Lena types questions to a study helper and decides what to write. What is it?

Answer: C. A chatbot

Conversation; she acts.

- ✗ A. A fixed workflow: no fixed steps
- ✗ B. A spreadsheet: no
- ✗ D. An agent: she decides

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F1-Q2 · question page 1987

AI.AGENT.01 · Classify AI setups in practice

Every night, a system collects new orders, uses a model to tag them, and files them. What is it?

Answer: D. A fixed workflow

Same steps each time.

- ✗ A. A chatbot: no conversation
- ✗ B. A calculator: no
- ✗ C. An agent: steps are fixed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F1-Q3 · question page 1988

AI.AGENT.01 · Classify AI setups in practice

A system told "book the cheapest flight to Denver next week" searches, compares and books. What is it?

Answer: C. An agent

Goal, tools, chosen steps.

- ✗ A. A search engine: it also books
- ✗ B. A fixed workflow: it chooses steps
- ✗ D. A chatbot: it acts

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F1-Q4 · question page 1989

AI.AGENT.01 · Classify AI setups in practice

Which system can affect the world most directly?

Answer: D. An agent that can send and buy

Actions in the world.

- ✗ A. A printed manual: no
- ✗ B. A calculator: no
- ✗ C. A chatbot that only shows text answers on a screen for the person to

- read and decide: text only

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F1-Q5 · question page 1990

AI.AGENT.01 · Classify AI setups in practice

A workflow uses a model only to summarize; a person designed every other step. What decides the order of steps?

Answer: A. The person's design

Fixed design.

- ✗ B. Chance: no
- ✗ C. The customer: no
- ✗ D. The model, which rearranges the order of the steps every time the workflow is started: fixed

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F2-Q1 · question page 1991

AI.AGENT.01 · Look past labels to what a system can do

A product is marketed as an "agent" but always runs the same three steps. What is it really?

Answer: B. A fixed workflow

Look at behavior.

- ✗ A. A database: no
- ✗ C. A chatbot: no conversation
- ✗ D. An agent: labels are loose

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F2-Q2 · question page 1992

AI.AGENT.01 · Look past labels to what a system can do

Which question best reveals the risk of a system?

Answer: D. "Can it act without a person approving?"

Unapproved actions matter.

- ✗ A. "Is it new?": not the risk
- ✗ B. "Is it popular?": not the risk
- ✗ C. "Does it have a friendly name and a nice logo?": irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F2-Q3 · question page 1993

AI.AGENT.01 · Look past labels to what a system can do

A chatbot quietly searches the web before answering. Does that make it an agent?

Answer: A. Not necessarily; it may still only show text

Ask what it can do.

- ✗ B. No, chatbots cannot use tools: they can
- ✗ C. Yes, if it is fast: no
- ✗ D. Yes, because any system that uses a tool: not necessarily

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F2-Q4 · question page 1994

AI.AGENT.01 · Look past labels to what a system can do

Which ability needs the most limits?

Answer: B. Deleting files

Irreversible actions.

- ✗ A. Answering a quiz question: low risk

✗ C. Suggesting a title: low risk

✗ D. Showing a short text summary: low risk

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S95-F2-Q5 · question page 1995

AI.AGENT.01 · Look past labels to what a system can do

Who decides the next step tells you what?

Answer: C. How much checking the system needs

Decision-maker shapes oversight.

✗ A. Nothing important, since every AI system needs exactly the same amount of checking: it varies

✗ B. Its speed: no

✗ D. Its price: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-M-Q1 · question page 1997

AI.AGENT.02 · Describe an agent's goal, tools, plan, feedback and stopping condition

"Find three venues under 500 dollars for 40 people" is which part?

Answer: B. Goal

What it is trying to achieve.

✗ A. Tool, because finding venues is something that a search tool does for the agent: it is the aim

✗ C. Feedback: no

✗ D. Stopping condition: partly, but it is the goal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-M-Q2 · question page 1998

AI.AGENT.02 · Describe an agent's goal, tools, plan, feedback and stopping condition

Search, calendar and email are which part?

Answer: D. Tools

What it can use.

- ✗ A. Plan: no
- ✗ B. Feedback: no
- ✗ C. Goals: they are means

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-M-Q3 · question page 1999

AI.AGENT.02 · Describe an agent's goal, tools, plan, feedback and stopping condition

"A venue is full, so look again" is which part?

Answer: C. Feedback

Results guide next steps.

- ✗ A. Goal, because looking again for another venue is the main thing the agent wants to do: it is a response
- ✗ B. Tool: no
- ✗ D. Stopping condition: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-M-Q4 · question page 2000

AI.AGENT.02 · Describe an agent's goal, tools, plan, feedback and stopping condition

"Stop after 20 searches" is which part?

Answer: A. Stopping condition

When to stop.

- ✗ B. Goal: no
- ✗ C. Tool: no
- ✗ D. Plan, because twenty searches is the list of steps the agent chooses to take in order: it is a limit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-M-Q5 · question page 2001

AI.AGENT.02 · Describe an agent's goal, tools, plan, feedback and stopping condition

What can a missing stopping condition cause?

Answer: C. Wandering or repeating steps

No clear end.

- ✗ A. Fewer tools: no
- ✗ B. A shorter plan: no
- ✗ D. Faster and better results: risky

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-B1-Q1 · question page 2002

AI.AGENT.02 · Compare an agent's parts with sending someone grocery shopping

"Ingredients for tacos for 6" is which part?

Answer: B. Goal

The aim.

- ✗ A. Feedback: no
- ✗ C. Tool: no
- ✗ D. Plan: it is the aim

Arithmetic checked by code: not applicable. Code example run and its output checked: not

applicable. Sources: none (stable)

S96-B1-Q2 · question page 2003

AI.AGENT.02 · Compare an agent's parts with sending someone grocery shopping

Money and a car are which part?

Answer: D. Tools

What can be used.

- ✗ A. Stopping condition: no
- ✗ B. Goals, because the friend is trying to get money and a car during the trip: they are means
- ✗ C. Feedback: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-B1-Q3 · question page 2004

AI.AGENT.02 · Compare an agent's parts with sending someone grocery shopping

"Out of tortillas, so try the bakery" is which part?

Answer: A. Feedback

Results change the plan.

- ✗ B. Stopping condition, since running out of tortillas means the shopping is finished: it continues
- ✗ C. Goal: no
- ✗ D. Tool: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-B1-Q4 · question page 2005

AI.AGENT.02 · Compare an agent's parts with sending someone grocery shopping

"Come back after spending 40 dollars" is which part?

Answer: C. Stopping condition

A budget limit.

- ✗ A. Tool: no
- ✗ B. Feedback: no
- ✗ D. Goal, because spending forty dollars is the main reason for going shopping at all: it is a limit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-B1-Q5 · question page 2006

AI.AGENT.02 · Compare an agent's parts with sending someone grocery shopping

What happens without a stopping condition?

Answer: A. Your friend may keep searching too long

No clear end.

- ✗ B. The tacos taste better: no
- ✗ C. Nothing changes: it matters
- ✗ D. Your friend will always come back sooner: not likely

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F1-Q1 · question page 2007

AI.AGENT.02 · Describe and improve agent setups in practice

An agent is told "organize my inbox" with no other details. What is missing?

Answer: B. A clear goal and stopping condition

Vague goals wander.

- ✗ A. More tools, because an agent with access to more tools always understands vague goals: not the fix
- ✗ C. Nothing: it is vague

✗ D. A shorter name: irrelevant

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F1-Q2 · question page 2008

AI.AGENT.02 · Describe and improve agent setups in practice

Which stopping condition is clearest?

Answer: D. "Stop after sorting 200 emails or 10 minutes."

Measurable limits.

✗ A. "Stop sometime.": vague
✗ B. "Never stop.": no limit
✗ C. "Stop whenever you feel that the inbox looks nice.": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F1-Q3 · question page 2009

AI.AGENT.02 · Describe and improve agent setups in practice

An agent searching for flights finds one sold out and searches again. What part is at work?

Answer: A. Feedback

Results guide next steps.

✗ B. Stopping condition: no
✗ C. Goal, because a sold-out flight is a new goal that the agent must achieve next: it is a result
✗ D. Tool: no

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F1-Q4 · question page 2010

AI.AGENT.02 · Describe and improve agent setups in practice

An agent can use a calendar but not email. Which part does this describe?

Answer: C. Tools

What it can use.

- ✗ A. Goal: no
- ✗ B. Feedback: no
- ✗ D. Plan: it is access

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F1-Q5 · question page 2011

AI.AGENT.02 · Describe and improve agent setups in practice

Before a high-stakes task, what is a sensible request?

Answer: B. "Show me your plan before acting."

Review the plan first.

- ✗ A. "Act as fast as possible.": no review
- ✗ C. "Use every tool you can.": risky
- ✗ D. "Go ahead with whatever plan you think is best, and tell me about it next week.": no review

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F2-Q1 · question page 2012

AI.AGENT.02 · Prevent misread goals and runaway loops

An agent told "get the most sign-ups" sends ten emails to each person. What went wrong?

Answer: D. The goal had no limits

Add limits to goals.

- ✗ A. The sign-up form was wrong: not stated

✗ B. The emails were too short: not the issue

✗ C. The agent was broken: it followed the goal

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F2-Q2 · question page 2013

AI.AGENT.02 · Prevent misread goals and runaway loops

Which goal includes a useful limit?

Answer: C. "Invite club members, no more than one email each."

Bounded goal.

✗ A. "Invite as many people as you possibly can, using any method you think will work.": no limit

✗ B. "Do your best.": vague

✗ D. "Invite people.": vague

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)

S96-F2-Q3 · question page 2014

AI.AGENT.02 · Prevent misread goals and runaway loops

What should a stopping condition include?

Answer: A. A budget of time, steps or money

Hard limits.

✗ B. Only a friendly message: not a limit

✗ C. Nothing specific, since agents always know by themselves when it is the right time to stop: not reliable

✗ D. A longer goal: not a limit

Arithmetic checked by code: not applicable. Code example run and its output checked: not applicable. Sources: none (stable)